MEASURING THE COST OF LAYERWISE COARSE-TO-FINE SAMPLING IN LATTICE ϕ^4 THEORY

ABSTRACT

Modular coarse-to-fine samplers can be enlarged by freezing and reusing trained stages. This modularity creates a consistency problem when each resolution is trained against its own physical action: the inherited parent distribution is generally not the coarse marginal required by the finer target. A lower bound is derived for refinements that preserve the inherited coordinates. Reference ensembles confirm that this mismatch persists across three adjacent resolutions and cannot be removed by field rescaling. To measure the cost of escaping this bound, a density-evaluable Haar lifting stage is constructed for two-dimensional near-critical ϕ^4 theory. New detail coefficients are generated by an ultraviolet transport, and the inherited coarse coefficients are corrected by an infrared transport. For frozen parents, the incremental log-weight variance per created coefficient, δ^2, is introduced as a stage-local lift-cost diagnostic. A paired parent–output covariance check supports this interpretation in every tested arm. Across three adjacent lifts, δ^2 varies by only 4.1%, while ESS/N decreases by a factor of 160. The measured rate is 31–44 times the per-site log-weight variance of a direct flow trained under the same update and batch schedule. Ablations locate the dominant difficulty in relaxation of the inherited coarse distribution rather than generation of new details. Joint optimization improves selected physical observables but recovers little of the efficiency gap. Thus, the main bottleneck in the tested modular construction is the cost of reconciling adjacent physical targets. The resulting bound and diagnostic turn adjacent-resolution consistency into a measurable design requirement for modular hierarchical sampling.

1 Introduction

Sampling near a continuous phase transition is limited by critical slowing down. In this regime, the correlation length approaches the system size, and long-wavelength modes evolve slowly. The cost of obtaining effectively independent lattice configurations therefore rises rapidly with volume [1, 2]. Metropolis and hybrid Monte Carlo remain standard routes to asymptotically exact sampling [3, 4, 5]. Because correlated configurations are constructed sequentially, the loss of efficiency becomes consequential at large volume.

Neural generative samplers address this sequential bottleneck by learning global proposals for the Boltzmann distribution. Normalizing flows are particularly useful because a bijective transformation provides an evaluable proposal density. The same model can therefore be trained without target samples, generate configurations directly, and support importance-weighted estimation or an exact independence Metropolis correction [6, 7, 8]. Continuous normalizing flows (CNFs) retain this density information while parameterizing the transformation by an ordinary differential equation (ODE), rather than by a finite stack of invertible layers [9, 10]. Combining CNFs with resolution-independent neural operators further allows parameters learned on a smaller lattice to initialize optimization at a larger discretization [11]. Such transfer can reduce optimization cost, but it does not remove the central burden of a one-shot fine-lattice model: one high-dimensional transformation must reproduce system-scale correlations, local interactions and non-Gaussian fluctuations simultaneously. Thus, although flow-based sampling can remain effective in critical regimes, training cost and mode coverage remain central scaling concerns [12, 13, 14].

A different organization is provided by scale separation. In real-space renormalization-group constructions, long-distance structure is represented by coarse variables, and shorter-distance degrees of freedom are introduced by successive refinement [15]. The broader pattern of enlarging a coarse state and then refining or re-equilibrating it appears in multiscale lattice thermalization, multiscale invertible Bayesian models and molecular backmapping [16, 17, 18], and the same coarse-to-fine principle has been applied by modern multiscale generative samplers to scalar and gauge fields [19, 20, 21]. As the required coarse partitions, wavelet coordinates and conditional refinements

are already established, the question considered here is how these components can be combined when every lattice resolution is assigned its own physical action.

This assignment creates an inconsistency between adjacent levels. A sampler trained at size L approximates the physical target π_L, while in general, this distribution differs from the low-pass marginal induced by the finer target π_{2L}. Consequently, the discrepancy cannot be removed by a refinement that preserves the inherited coarse variables, which remains true even if the conditional distribution of every newly introduced detail is represented exactly. In Section 2.3.2, the obstruction is stated as a lower bound on the refinement error, and therefore, the resulting marginal mismatch is the main subject of this work.

Two approaches can be used to avoid the bound, with different trade-offs. In the first, the full hierarchy is trained end to end against the finest action. No intermediate physical action is then imposed, and each coarse level can learn the marginal required by the finest target, so the obstruction just does not arise. This approach is followed by recent multiscale lattice samplers, for which high sampling efficiency has been reported [21]. Modularity is reduced, however, because the hierarchy is tied to a single final lattice size. A larger volume therefore requires the full stack to be retrained, and all levels must be retained in the optimization graph.

In the second approach, the levels are kept separate, and each transition is supervised with its own physical action. An explicit mechanism is provided at each stage to correct the inherited coarse marginal. Once accepted, a stage is frozen and reused, so modular growth is preserved. The required correction must then be learned by a coordinate-restricted transport acting on a frozen parent. This second approach is studied here, and its cost is quantified.

Each $L \to 2L$ transition is decomposed into IR and UV coordinates by a fixed Haar transform. The newly resolved detail coefficients are initialized with a physics-informed conditional Gaussian. Their remaining fine-scale structure is modeled by a UV transport, while the inherited low-pass distribution is corrected by a separate IR transport. Both restricted transports are trained jointly against the physical action at size $2L$. The completed stage is then frozen and used as the parent of the next resolution. In this way, a tractable composed density is retained without backpropagation through the full multilevel hierarchy.

Freezing a completed stage preserves modularity, but it also propagates any error in its parent. In practice, the frozen model q_L need not equal the physical target π_L. Every reported lift is trained from a fixed finite pool drawn from q_L, with parents selected uniformly and without importance resampling. The saved hierarchy and all evaluations use the same unfiltered parent law, so the reported results characterize the actual composed model rather than an idealized lift supplied with exact coarse samples. Appendix F analyzes a more general target-corrected selection law and explains why it cannot remove the adjacent-resolution inconsistency studied here.

The **main contributions** are as follow:

- An adjacent-resolution consistency obstruction is identified for modular hierarchies whose levels are supervised by different physical actions. A lower bound shows that a refinement preserving inherited coarse coordinates cannot remove a mismatch with the coarse marginal of the finer target, therefore, A rescaling-invariant reference-ensemble test is provided to determine whether this obstruction is active.
- A density-evaluable layerwise construction is introduced to avoid the bound while retaining modularity, with a Haar decomposition assigns newly resolved details and inherited coarse coefficients to separate UV and IR transports. Accepted stages can be frozen and reused without backpropagation through the full hierarchy.
- The stage-local diagnostic δ^2 is defined for frozen-parent lifts. It measures incremental log-weight variance per created degree of freedom and separates the cost of a lift from dimensional growth. A paired parent–output decomposition quantifies the covariance term and tests this interpretation directly.
- Layerwise, jointly optimized and single-scale flows are compared under controlled training and evaluation protocols, while frozen-transport, parent-law and conditioning ablations are used to locate the measured cost. Relaxation of the inherited coarse marginal is identified as the dominant challenge in the tested factorization. The comparisons also show why agreement in selected observables and full-density sampling diagnostics must be assessed separately.
- The measured cost is converted into a scaling criterion for a modular hierarchy: a lifting stage becomes competitive when its variance cost per created coefficient approaches the per-site cost of a direct flow. This criterion makes the accumulation of stage error explicit and quantifies the improvement required before a deeper hierarchy becomes useful.

The target, the consistency problem and its measured magnitude are defined in Section 2. The construction is positioned relative to sample-free, image-domain and renormalization-group-informed multiscale models in Section 3. The

method is presented in Section 4, and the experimental setup is described in Section 5. The measurements and their interpretation are given in Sections 6 and 7, respectively.

2 Preliminaries

The target lattice field theory and the density-evaluable continuous transports used in this work are introduced first. The motivating multiresolution consistency problem is then defined and measured on reference ensembles.

2.1 The Two-Dimensional ϕ^4 Lattice Target

A real scalar field is considered on a periodic two-dimensional square lattice,

$$\Lambda_L = \{1, \dots, L\}^2, \qquad \phi = \{\phi_x \in \mathbb{R} : x \in \Lambda_L\}. \tag{1}$$

The L^2 real components of ϕ are the degrees of freedom (DOF) of the target at size L. This count is preserved under every invertible change of coordinates on ϕ. In the mass-quartic parametrization, the dimensionless lattice action is

$$S_L(\phi) = \sum_{x \in \Lambda_L} \left[\frac{1}{2} \sum_{\mu=1}^{2} (\phi_{x+\hat{\mu}} - \phi_x)^2 + \frac{m^2}{2} \phi_x^2 + \frac{\lambda}{4} \phi_x^4 \right], \tag{2}$$

where all indices are interpreted periodically. The nearest-neighbor term penalizes rapid spatial variation, while the local potential is controlled by m^2 and $\lambda > 0$. The action is invariant under the global $\mathbb{Z}_2$ transformation $\phi \mapsto -\phi$, and the corresponding target is

$$\pi_L(\phi) = \frac{1}{Z_L} \exp[-S_L(\phi)], \qquad Z_L = \int \exp[-S_L(\phi)] \, d\phi, \tag{3}$$

where the partition function Z_L is generally intractable.

The following couplings are used in the experiments:

$$m^2 = -0.4658, \qquad \lambda = 0.3007, \tag{4}$$

These values are obtained by converting the near-critical parameters $\kappa_c = 0.2705$ and $\widetilde{\lambda} = 0.022$ used in previous work [21]. The action convention of Eq. 2 and the field rescaling $\phi = \sqrt{2\kappa}\,\widetilde{\phi}$ are used in the conversion.

Near a continuous phase transition, the correlation length becomes comparable to the lattice size. Slowly evolving long-wavelength modes then cause the integrated autocorrelation time of local Markov-chain updates to grow approximately as

$$\tau_{\mathrm{int}} \propto L^{z_{\mathrm{dyn}}}, \tag{5}$$

where z_{dyn} is the dynamical critical exponent. This critical slowing down motivates samplers that represent long-distance and short-distance structure at different spatial resolutions [1, 2, 13].

The field is summarized by the mean magnetization

$$m(\phi) = L^{-2} \sum_{x \in \Lambda_L} \phi_x, \tag{6}$$

from which a sector-width statistic and the fourth-order Binder cumulant are formed [22]:

$$U_4 = 1 - \frac{\langle m^4 \rangle}{3 \langle m^2 \rangle^2}. \tag{7}$$

The Binder cumulant is a dimensionless ratio and is therefore invariant under any rescaling $\phi \mapsto \alpha\phi$ with $\alpha \neq 0$. This property is used in Section 2.4 to isolate the part of a coarse-graining correction that no linear map can reproduce.

2.2 Continuous Normalizing Flows for Unnormalized Targets

Let p_0 be a tractable source distribution, with endpoint target $p_1 = \pi_L$. A sample is evolved by a deterministic continuous transport:

$$\frac{dX_t}{dt} = v_\theta(X_t, t), \qquad X_0 \sim p_0, \tag{8}$$

where v_θ is a parameterized velocity field. An invertible endpoint map is defined by integration from $t = 0$ to $t = 1$, with its endpoint model distribution is denoted by q_θ [9, 10].

2.2.1 Density Evolution

Along a continuous trajectory, the log density satisfies the instantaneous change-of-variables equation

$$\frac{d}{dt}\log p_t(X_t) = -\nabla \cdot v_\theta(X_t, t). \tag{9}$$

Consequently,

$$\log q_\theta(X_1) = \log p_0(X_0) - \int_0^1 \nabla \cdot v_\theta(X_t, t)\, dt. \tag{10}$$

For a high-dimensional state, the divergence may be evaluated analytically for a suitably structured velocity field or estimated using the Hutchinson trace estimator,

$$\nabla \cdot v_\theta = \operatorname{tr}\left(\frac{\partial v_\theta}{\partial X_t}\right) \approx \epsilon^\top \frac{\partial v_\theta}{\partial X_t}\epsilon, \qquad \mathbb{E}\left[\epsilon\epsilon^\top\right] = I. \tag{11}$$

The implemented density remains subject to the numerical error of the ODE solver and of the associated density integration.

2.2.2 Reverse-KL Training and Importance Weights

For the normalized target $\pi_L(X) = Z_L^{-1}\exp[-S_L(X)]$, the reverse Kullback-Leibler (KL) divergence is

$$D_{\mathrm{KL}}(q_\theta \| \pi_L) = \mathbb{E}_{X\sim q_\theta}\left[\log q_\theta(X) - \log \pi_L(X)\right] \tag{12}$$

$$= \mathbb{E}_{X\sim q_\theta}\left[\log q_\theta(X) + S_L(X)\right] + \log Z_L. \tag{13}$$

Because Z_L is independent of θ, the trainable loss is

$$\mathcal{L}_{\mathrm{RKL}}(\theta) = \mathbb{E}_{X\sim q_\theta}\left[S_L(X) + \log q_\theta(X)\right]. \tag{14}$$

This loss requires samples from the current model and evaluations of the target action, but no samples from the target distribution.

An unnormalized endpoint importance weight is

$$w(X) = \frac{\exp[-S_L(X)]}{q_\theta(X)}, \qquad \log w(X) = -S_L(X) - \log q_\theta(X). \tag{15}$$

These weights enable self-normalized importance sampling and provide a density-based effective sample size (ESS) diagnostic [7, 8].

Two properties of $\log w$ are used throughout. First, the identity $\log w = \log(\pi_L/q_\theta) + \log Z_L$ holds, with $\log Z_L$ constant. The log-weight variance is therefore

$$\sigma^2_{\log w} = \mathrm{Var}_{q_\theta}\left[\log \frac{\pi_L}{q_\theta}\right]. \tag{16}$$

Up to a sign, and a conventional factor of one half, this is the log-variance loss used in variational inference and the variance of relative surprisal studied in information theory [23, 24]. It vanishes if and only if $\log(\pi_L/q_\theta)$ is constant q_θ, which is a general case. For the full-support models used here, the normalized densities then satisfy $q_\theta = \pi_L$ almost everywhere. Thus, the log-weight variance vanishes only when the target distribution is reached.

Second, the self-normalized ESS is related to the χ^2 divergence, whose equivalent second-weight-moment form is a standard control on generic importance-sampling error [25]. From $\mathbb{E}_{q_\theta}[w] = Z_L$ and $\mathbb{E}_{q_\theta}[w^2] = Z_L^2 \int \pi_L^2/q_\theta$, its large-$N$ limit is

$$\frac{\mathrm{ESS}}{N} \longrightarrow \frac{\left(\mathbb{E}_{q_\theta}[w]\right)^2}{\mathbb{E}_{q_\theta}[w^2]} = \frac{1}{1+\chi^2(\pi_L \,\|\, q_\theta)}, \qquad \chi^2(\pi_L \,\|\, q_\theta) = \int \frac{\pi_L^2}{q_\theta} - 1, \tag{17}$$

This relation is exact when the second weight moment is finite, and no assumption about the shape of the weight distribution is required. For approximately log-normal weights, the normalized ESS behaves as

$$\frac{\mathrm{ESS}}{N} \approx \exp\left(-\sigma^2_{\log w}\right), \tag{18}$$

Thus, a fixed per-site error in $\log w$ produces an exponential decay of efficiency with lattice volume. For a model transferred between volumes, the same statement is written in Ref. [26] as $\mathrm{ESS}(V) = \mathrm{ESS}(V_0)^{V/V_0}$, which is Eq. 18 at fixed $\sigma^2_{\log w}$ per site. Comparisons between models at different lattice sizes can therefore be reported per site.

2.3 Multiresolution Consistency

2.3.1 Coarse-Graining and Conditional Factorization

Let x_ℓ denote a configuration at resolution ℓ, with physical target

$$\pi_\ell(x_\ell) = \frac{1}{Z_\ell} \exp[-S_\ell(x_\ell)]. \tag{19}$$

A coarse-graining map R_ℓ extracts a lower-dimensional coordinate

$$c = R_\ell(x_\ell). \tag{20}$$

Let d contain the remaining DOF so that (c, d) is a complete invertible representation of x_ℓ. The target then factorizes as

$$\pi_\ell(c, d) = \pi_\ell^{\mathrm{IR}}(c)\pi_\ell^{\mathrm{UV}}(d \mid c). \tag{21}$$

The induced coarse marginal can be written formally as

$$\pi_\ell^{\mathrm{IR}}(c) = \int \pi_\ell(x_\ell)\delta\left(c - R_\ell(x_\ell)\right)\, dx_\ell. \tag{22}$$

Integrating out the unresolved coordinates generally introduces nonlocal and higher-order interactions, even when S_ℓ is local. Neither the exact coarse marginal nor the exact conditional distribution is therefore available in closed form for the interacting target.

2.3.2 Adjacent-Resolution Marginal Mismatch

Consider physical targets $\pi_{\ell-1}$ and π_ℓ at two adjacent lattice sizes. Let E_ℓ map a configuration from resolution $\ell - 1$ into the coarse coordinates used at resolution ℓ,

$$c = E_\ell(x_{\ell-1}). \tag{23}$$

Even if the parent sampler exactly represents $\pi_{\ell-1}$, its inherited coarse distribution is the pushforward

$$\overline{\pi}_{\ell-1} = (E_\ell)_{\#}\, \pi_{\ell-1}. \tag{24}$$

A refinement that preserves c and generates only the unresolved coordinates has distribution

$$q_\ell(c, d) = \overline{\pi}_{\ell-1}(c)q_\ell(d \mid c), \tag{25}$$

whereas the finer physical target factorizes as

$$\pi_\ell(c, d) = \pi_\ell^{\mathrm{IR}}(c)\pi_\ell^{\mathrm{UV}}(d \mid c). \tag{26}$$

The chain rule for the KL divergence gives

$$\begin{aligned} D_{\mathrm{KL}}\left(q_\ell \| \pi_\ell\right) = {} & D_{\mathrm{KL}}\left(\overline{\pi}_{\ell-1} \| \pi_\ell^{\mathrm{IR}}\right) \\ & + \mathbb{E}_{c\sim\overline{\pi}_{\ell-1}}\left[D_{\mathrm{KL}}\left(q_\ell(d \mid c) \| \pi_\ell^{\mathrm{UV}}(d \mid c)\right)\right]. \end{aligned} \tag{27}$$

Consequently,

$$D_{\mathrm{KL}}\left(q_\ell \| \pi_\ell\right) \geq D_{\mathrm{KL}}\left(\overline{\pi}_{\ell-1} \| \pi_\ell^{\mathrm{IR}}\right). \tag{28}$$

Exact conditional generation removes the second term but cannot remove the first whenever

$$\overline{\pi}_{\ell-1} \neq \pi_\ell^{\mathrm{IR}}. \tag{29}$$

This obstruction exists even for an exact parent sampler. In practice, an additional discrepancy is introduced by approximation error in the parent model. A stage supervised directly with the finer physical action must therefore perform two tasks: the inherited infrared (IR) coordinates must be modified, and the unresolved ultraviolet (UV) coordinates must be generated.

The scope of the bound should be noted. It applies only to refinements that leave c unchanged. If c is instead treated as a latent variable that may be transported, a bijection can be formed between the parent variables and the fine configuration. The bound then no longer applies. This observation motivates the IR transport in Section 4, through which Eq. 28 is avoided. The size of the required correction is examined next.

2.4 Is the Required Correction a Rescaling?

Equation 28 shows that a refinement preserving the coarse coordinates is bounded away from the target, but does not specify the form of the correction needed to remove the bound. If the inherited and required coarse ensembles differ only by a multiplicative field rescaling, one constant per level is sufficient, and no learned coarse transport would then be needed. If their shapes differ, however, a constant is insufficient and a flow is required.

This question is resolved by measurements on reference Markov chain ensembles at the couplings of Eq. 4. No learned model is involved, and no sampling result of this work is reported in this subsection.

Prediction of renormalization-group scaling. The non-overlapping 2×2 block average is used as the coarse coordinate. Up to a constant, it coincides with the Haar low-pass coefficient. At an exact critical fixed point, the theory is scale invariant. Blocking a configuration of the $2L$ theory should then reproduce the L theory up to one multiplicative field rescaling. The corresponding exponent is fixed by the anomalous dimension. Away from the fixed point, this picture is modified by corrections to scaling [27, 28]. These corrections are strongest at small lattice sizes. Whether the remaining discrepancy requires a flow is therefore an empirical question.

A rescaling-invariant diagnostic. For each adjacent pair, the bare ensemble at size L is compared with the block-averaged ensemble obtained at size $2L$. These ensembles are sample-based realizations of $\overline{\pi}_L$ and π^{IR}_{2L}, respectively. The Binder cumulant of Eq. 7 is invariant under $\phi \mapsto \alpha\phi$. Therefore, the difference

$$\Delta U_4 = U_4\big(\pi^{\mathrm{IR}}_{2L}\big) - U_4(\overline{\pi}_L) \tag{30}$$

vanishes for every pure rescaling, regardless of the value of α. A nonzero ΔU_4 therefore shows that part of the required correction cannot be reproduced by rescaling. Two amplitude diagnostics are also included in Table 1: the single-site standard-deviation ratio r_σ and the mean-squared-magnetization ratio r_{m^2}. Both respond to a rescaling and are analyzed in Appendix E. Each ensemble contains 10^4 configurations, except at $L = 64$, where 2×10^4 are used. All quantities follow the convention of Eq. 2.

Table 1: Correction required of the coarse transport at each resolution transition. The bare ensemble at size L is compared with the block-averaged ensemble from size $2L$. The rescaling-invariant quantity ΔU_4 isolates the component that no rescaling can reproduce. The quantities r_σ and r_{m^2} respond to rescaling and are discussed in Appendix E.

transition	ΔU_4	r_σ	r_{m^2}
$L = 4$ from $L = 8$	$+0.0642$	0.8427	0.7424
$L = 8$ from $L = 16$	$+0.0365$	0.8510	0.7568
$L = 16$ from $L = 32$	$+0.0149$	0.8563	0.7658
$L = 32$ from $L = 64$	-0.0149	0.8575	0.7128

A block jackknife is applied to the reference chains, with blocks of 50 stored configurations. Standard errors of 0.0041, 0.0033, 0.0036 and 0.0048 are obtained for the four values of ΔU_4 in table order. The need for more than a rescaling is therefore statistically resolved at the first three transitions. A separate systematic reservation about the $L = 64$ reference is discussed in Section 7.7.

Outcome. ΔU_4 is statistically nonzero at the three primary transitions. No choice of α therefore makes the inherited coarse ensemble match the required one. A constant per level is insufficient, and the shape of the coarse distribution must be changed by a transport. For this reason, an explicit IR flow is included in the stage of Section 4.

The scalar witness decreases from 0.0642 at the coarsest transition to 0.0149 at $16 \to 32$, which is a factor of 4.3. Because it can be measured before any model is trained, ΔU_4 is a candidate explanation for the cost of a lifting stage. If it governs that cost, refinement should become cheaper on larger lattices. This possibility is treated as a hypothesis and is tested against the measured cost in Section 7.3.

Caveats. The underlying physics is standard finite-size scaling [28, 27, 22], and no new result is claimed in that respect. As an ensemble check, the bare Binder cumulant at $L = 32$ is measured as 0.5689, and agreement is found with the value reported independently at the same coupling [21].

The measurement characterizes the correction required of the coarse transport, meaning it cannot, by itself, predict the sampling efficiency that will be achieved. Model capacity and optimization also contribute, and these effects are separated in Section 6.

3 Related Work

3.1 Target-Sample-Free Neural Sampling

When only an unnormalized action is available, a neural sampler must be trained without configurations drawn from the target, as in reverse-KL normalizing flows, whose training requires only samples from the current model and evaluations of the action [6]. Because their endpoint densities are evaluable, these models also support importance weighting, asymptotically unbiased estimators and exact independence Metropolis corrections [7, 8]. This framework has been extended to gauge fields through symmetry-preserving architectures [29], and although studies of critical models have demonstrated successful sampling, they also identify training cost and mode coverage as unresolved scaling concerns [12, 13, 14].

For continuous transformations, neural ODEs and FFJORD provide the standard CNF density machinery [9, 10], while neural-operator CNFs extend this construction across lattice discretizations and show that pretraining at smaller sizes can accelerate optimization on a larger lattice [11].

In other sample-free methods, the action is used to guide the transport in different ways: iterated denoising energy matching alternates model sampling with a simulation-free denoising objective based on the energy and its gradient [30], whereas the non-equilibrium transport sampler augments a finite-time Langevin quench with a learned drift and evolves correction weights along the stochastic path [31]. Flow Annealed Importance Sampling Bootstrap uses annealed importance sampling to expose underrepresented regions before optimizing a mass-covering flow objective [32], while Controlled Monte Carlo Diffusions learn the forward and backward dynamics of an annealed stochastic path [33].

Because these methods provide different forms of density or path-weight information, they require method-specific diagnostics, and their objectives, although relevant as experimental comparisons, are distinct from the spatial decomposition studied here. The present implementation uses reverse-KL CNFs, with the contribution lying in how a target-sample-free transformation is organized across resolutions.

3.2 Log-Weight Diagnostics and Dimension Scaling

Importance-sampling performance is conventionally characterized through the second moment of the weights, or equivalently through the χ^2 divergence and its ESS proxy [25, 26]. The variance of the log density ratio is also an established quantity: it is used as the log-variance loss in variational inference [23] and as the variance of relative surprisal in information theory [24]. With samples drawn from the proposal, these quantities coincide, up to sign and normalization conventions, with $\sigma^2_{\log w}$ in Eq. 16.

The scaling of weight fluctuations with dimension is a recurring concern: particle-filter analyses relate the ensemble size required to avoid collapse to the variance of the observation log likelihood [34], annealed importance sampling for the Ising model has been assessed through the empirical variance of log normalized weights along its annealing path [35], and lattice-flow studies likewise use log-weight variance per DOF to compare volumes [21]. These precedents motivate reporting $\sigma^2_{\log w}/L^2$, but they do not provide the stage-local accounting used in Section 5.3: δ^2 subtracts the inherited parent variance before normalizing by the number of coordinates created by a lift and, because the paired child and parent weights are correlated, is neither a standard divergence nor simply the variance of a log-weight increment.

3.3 Multiresolution Generative Modeling from Data

Conditional coarse-to-fine generation is well established in image modeling. SRFlow learns the conditional distribution of plausible high-resolution images given a low-resolution observation [36], Wavelet Flow factorizes an image distribution into a low-resolution model and wavelet-detail conditionals with tractable likelihoods [37], and multi-resolution CNFs characterize the additional information required at each finer resolution before combining it with the coarser image through a likelihood-preserving linear map [38].

A closer cross-domain analogue to repeated lifting is provided by Multiscale Invertible Generative Networks, in which MsIGN upsamples the current state at each scale, augments it with latent variables and refines the enlarged state with an invertible network for high-dimensional Bayesian inference [17]. Progressive deblurring diffusion models instead begin from a low-frequency image and restore higher-frequency content during reverse diffusion [39]; because their state dimension remains fixed, they provide a frequency-by-frequency refinement analogy rather than a literal dimension-increasing lift.

Although conditional refinement and wavelet-based multiresolution flows are therefore established, their statistical setting differs from the one considered here. In image applications, the coarse image is obtained from, or paired with, a high-resolution sample, so the learned coarse distribution and fine conditional are induced by the same data-generating

distribution and the relation in Section 2.3.2 holds by construction. In the present lattice setting, by contrast, no target configurations are used, and π_L and π_{2L} are specified separately by their physical actions; their relation therefore need not coincide with the downsampling relation of paired image data.

Molecular backmapping provides a second generate-and-relax precedent. In hierarchical polymer reconstruction, equilibrated soft-blob configurations are resolved stepwise to mesoscopic and atomistic detail, and the system is relaxed after each reintroduction of DOF [18], while DeepBackmap learns the reverse map from equilibrated coarse molecular structures to local atomistic environments [40]. Both preserve coarse structural information while reconstructing discarded detail, but they rely on simulation data and subsequent physical relaxation rather than an exactly density-evaluable, target-sample-free transformation.

3.4 RG-Informed Multiscale Lattice Sampling

The physical motivation for separating length scales is provided by the renormalization group (RG), including the Kadanoff scaling picture [15], while multigrid Monte Carlo established that updates organized by length scale can address slow long-wavelength modes [2]. The Wavelet Conditional Renormalization Group makes this connection explicit by estimating scale-conditional energy models from observed physical configurations in an orthogonal wavelet basis and generating new fields from coarse to fine [41]; it thereby demonstrates the value of wavelet coordinates near phase transitions, but relies on a dataset rather than sampling from a known unnormalized action without target data.

An earlier lattice realization of the generate-and-relax pattern is multiscale Monte Carlo equilibration. In pure Yang–Mills theory, configurations sampled from an RG-matched coarse action are prolongated to a fine lattice and then rethermalized with conventional fine-lattice updates [16], and the construction was subsequently tested in two-color quantum chromodynamics (QCD) with dynamical fermions [42]. These methods use the coarse state to initialize the slow long-distance modes while local fine dynamics repair short-distance artifacts, making them direct algorithmic precedents for generating new fine variables and relaxing old ones, although their relaxation is a Markov-chain equilibration rather than a learned invertible map with an explicit proposal density.

Multiscale normalizing flows have also been constructed directly for lattice theories: hierarchical gauge-field flows build configurations from coarse to fine so that different components operate at different physical scales [19], whereas super-resolving normalizing flows upsample a coarse field, add constrained ultraviolet noise and apply a CNF to the entire fine field, with the option of optimizing the coarse source distribution [43]. The latter is the closest learned architectural precedent for an augment-and-relax lift, although its full-field CNF does not enforce the coordinate-triangular UV/IR separation used here. Physics-informed RG kernels provide another layerwise construction with independently optimizable propagation kernels [44], while near criticality, RiGCS generates Ising configurations hierarchically using approximate scale invariance and learns residual interactions when finite-size effects violate the idealized scale-invariant conditional model [20].

The closest target-level comparison is a recent multiscale sampler for ϕ^4 theory [21], which combines a conditional Gaussian mixture with masked CNFs and preserves the coarse sites exactly during refinement, thereby providing the exact restriction maps required for multilevel Monte Carlo. Preserving the coarse coordinates is also the condition under which Eq. 28 applies.

The bound is nevertheless avoided because no physical action is imposed at the intermediate resolutions: the full hierarchy is optimized end to end against the finest action alone, allowing its coarse levels to learn the marginals π_ℓ^{IR} instead of the separately defined targets π_ℓ. Exact restriction and end-to-end optimization are therefore complementary parts of this design, with the latter removing the obstruction.

The factorization of Section 2.3.1 therefore admits two classes of solution: the correct coarse marginals can be learned implicitly by optimizing every level against the finest action, or levelwise supervision can be retained while the inherited marginal is corrected explicitly. The first route, used by end-to-end multiscale samplers, can achieve high efficiency but couples the hierarchy to one final lattice size, so growth to a larger volume requires retraining the stack while retaining every level in one optimization graph. The second route preserves modular growth by freezing, reusing and composing accepted stages, but in return must learn the correction from a frozen parent through a coordinate-restricted transformation.

Taken together, this literature establishes the broad generate-and-relax pattern and its individual ingredients, including Kadanoff-inspired partitioning, wavelet coordinates, conditional refinement, rethermalization and layerwise training. The distinction pursued here is to combine them in a density-evaluable, coordinate-triangular lift under frozen, levelwise physical supervision, quantify the required inherited-marginal correction, and compare its cost with direct single-scale training and the tested joint-training alternatives.

4 Wavelet-Structured Layerwise UV/IR Transport

A hierarchy consists of a base model followed by lifting stages, each of which carries the sampler from L to $2L$. The base model is an unconditional CNF of the type described in Section 2.2, trained directly with the physical action at the coarsest size L_0, while each subsequent stage is constructed as follows.

The lattice is partitioned into Kadanoff 2×2 blocks and expressed in Haar coordinates through an orthonormal, exactly invertible change of variables that separates one coarse field from three detail fields. The parent therefore occupies the coarse coordinates exactly, with its embedding contributing only a constant to the log density.

The newly resolved detail coefficients are drawn from the conditional free theory associated with the same ϕ^4 action, which supplies in closed form the leading dependence of the details on the coarse field. Because this dependence is also present in the interacting theory, each stage begins from a physically motivated distribution.

Two coordinate-restricted transformations are then applied: the UV transformation moves the details at fixed coarse coordinates, whereas the IR transformation moves the coarse coordinates at fixed details. The latter is required because every refinement that preserves the coarse coordinates is bounded below by Eq. 28.

Both transformations use a velocity field with closed-form divergence, so the density of the composed stage is exact up to numerical integration of the flow; they are trained with the physical action at size $2L$, while every previously accepted stage remains frozen.

Let the represented lattice sizes be

$$L_\ell = 2^\ell L_0, \qquad \ell \in \{0, \dots, K\}. \tag{31}$$

Once accepted, the base model q_{L_0} is frozen, and each subsequent stage maps a parent field $X_L \sim q_L$ to a field X_{2L} at the next physical resolution.

One stage introduces independent detail noise $\epsilon \sim \mathcal{N}(0, I_{3L^2})$ and applies

$$X_L \xrightarrow{c_0 = 2X_L} c_0, (c_0, \epsilon) \xrightarrow{d_0 \sim \pi_{\text{free}}(\cdot \mid c_0)} (c_0, d_0) \xrightarrow{T_{\text{UV}}} (c_0, d_1) \xrightarrow{T_{\text{IR}}} (c_1, d_1) \xrightarrow{W_{\text{IRUV}}^{-1}} X_{2L}. \tag{32}$$

Here, c denotes one low-pass IR coefficient field, and $d = (d_{\text{H}}, d_{\text{V}}, d_{\text{D}})$ denotes the three UV detail fields; only d is changed by the UV transformation, and only c by the IR transformation. Both are updated in one optimization against the physical action S_{2L}, after which the complete sampler q_{2L} is frozen and used as the parent at the next resolution.

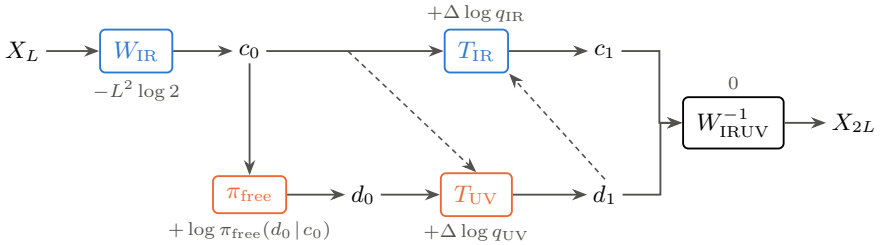


Figure 1: One resolution stage, $L \to 2L$, with the accepted parent X_L entering on the left and the generated fine configuration X_{2L} leaving on the right. Solid arrows carry state, whereas dashed arrows carry only conditioning. The parent is embedded in the low-pass coefficient c_0, and the details d_0 are drawn from the conditional free theory of Section 4.2 before being moved at fixed c_0 by the UV transformation. Conditioned on the *generated* details d_1, the IR transformation then moves the coarse coefficients. The oppositely directed dashed arrows represent sequential triangular maps whose determinant contributions multiply, as proved in Appendix D. Read from left to right, the annotations reconstruct Eq. 50; the orthonormal inverse Haar transform makes no additional contribution.

4.1 Layer Partition: Kadanoff Blocks and Haar Coordinates

At each resolution change, the non-overlapping 2×2 spatial partition of Kadanoff blocking fixes how neighboring fine-lattice sites are grouped [15]. For $i, j = 0, \dots, L-1$, the corresponding block vector is

$$\mathbf{x}_{ij} := \begin{pmatrix} X_{2i,2j} & X_{2i,2j+1} & X_{2i+1,2j} & X_{2i+1,2j+1} \end{pmatrix}^\top, \tag{33}$$

with lattice indices interpreted periodically. The four variables in each block are resolved by the orthonormal Haar basis into one low-pass IR coefficient and three high-pass UV coefficients, with the partition supplying the spatial grouping and the basis supplying an invertible coordinate decomposition that retains every fine-lattice DOF. Because the target at

resolution L is defined by the physical action at that size rather than by the block marginal of the target at $2L$, the two distributions differ as measured in Table 1.

For a fine field X_{2L}, the resulting fixed one-level Haar transform is

$$(c, d) = W_{\mathrm{IRUV}} X_{2L}. \tag{34}$$

Within each 2×2 block, its action is

$$\begin{pmatrix} c \\ d_{\mathrm{H}} \\ d_{\mathrm{V}} \\ d_{\mathrm{D}} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_{00} \\ x_{01} \\ x_{10} \\ x_{11} \end{pmatrix}. \tag{35}$$

Because the transform is orthonormal, its log absolute determinant is zero, and the coefficient shapes for one stage are

$$c \in \mathbb{R}^{L \times L}, \qquad d \in \mathbb{R}^{3 \times L \times L}. \tag{36}$$

Before a complete fine field exists, the accepted parent is embedded directly into the initial IR coefficients using

$$c_0 = W_{\mathrm{IR}}(X_L) = 2X_L. \tag{37}$$

The factor of two is fixed by the orthonormal Haar convention, which gives the low-pass coefficient obtained by repeating each parent value over a 2×2 block, and the parent-to-IR map therefore has

$$\log \left| \det \frac{\partial c_0}{\partial X_L} \right| = L^2 \log 2. \tag{38}$$

4.2 New-Site Initialization: Conditional Free Theory

The initial details are drawn from a normalized conditional Gaussian obtained from a positive-definite free theory. Let

$$S_{\mathrm{free}}(X) = \frac{1}{2} X^\top A X, \qquad K = W_{\mathrm{IRUV}} A W_{\mathrm{IRUV}}^\top = \begin{pmatrix} K_{cc} & K_{cd} \\ K_{dc} & K_{dd} \end{pmatrix}. \tag{39}$$

With the numerical stabilization $\overline{K}_{dd} = K_{dd} + \varepsilon_K I$, the implemented conditional source is

$$\pi_{\mathrm{free}}(d \mid c) = \mathcal{N}\left(-\overline{K}_{dd}^{-1} K_{dc} c,\ \overline{K}_{dd}^{-1} \right). \tag{40}$$

The source is evaluated in the coarse Fourier basis with independent 3×3 detail-precision solves at each momentum and is fixed throughout training, providing a normalized, spatially correlated UV initialization.

This choice reflects the physics of the detail coefficients, which contain the high-momentum content of the fine lattice and therefore lie in a regime where the kinetic term dominates the local potential and the conditional distribution is close to Gaussian. For the quadratic action, the conditional mean in Eq. 40 is linear in c, while the conditional covariance is independent of c, so the UV transformation begins with the leading coarse-field dependence already present and needs only to represent the remaining difference from the interacting target. That difference includes both the mismatch between the positive source mass and the physical quadratic term and the quartic interaction.

4.3 Transports: UV Generation of New Sites and IR Relaxation of Old Sites

The UV CNF evolves only the detail coefficients,

$$\frac{dd_t}{dt} = v_{\mathrm{UV},\theta}(d_t, t; c_0), \qquad d_{t=0} = d_0, \qquad d_{t=1} = d_1, \tag{41}$$

while c_0 remains fixed. Its log-density contribution is

$$\Delta \log q_{\mathrm{UV}} = -\int_0^1 \nabla_d \cdot v_{\mathrm{UV},\theta}(d_t, t; c_0)\, dt. \tag{42}$$

The IR CNF then evolves only the inherited low-pass coefficients and conditions on the generated details,

$$\frac{dc_t}{dt} = v_{\mathrm{IR},\varphi}(c_t, t; d_1), \qquad c_{t=0} = c_0, \qquad c_{t=1} = c_1, \tag{43}$$

while d_1 remains fixed. Its contribution is

$$\Delta \log q_{\mathrm{IR}} = -\int_0^1 \nabla_c \cdot v_{\mathrm{IR},\varphi}(c_t, t; d_1)\, dt. \tag{44}$$

As discussed in Section 4.2, the leading dependence of the detail distribution on the coarse field is a displacement of the conditional mean at fixed covariance, so the UV source is translated as a function of c_0. Because Section 2.4 shows that a rescaling cannot correct the inherited coarse coordinates, the IR transformation must alter both the shape and location of the coarse distribution.

The two transformations are coordinate-restricted components of a single invertible stage, whose bijectivity is proved in Appendix D.

4.4 Velocity Field: Analytic Backbone with Additive Conditioning

The same translation-equivariant velocity field is used in both transports, and its divergence is available in closed form. For a transported coefficient field x with condition y, site-wise features are formed from a linear term and trainable sine frequencies:

$$F(x) = P\left[\sin(\omega_1 x), \ldots, \sin(\omega_{n-1} x),\, x\right]^\top, \tag{45}$$

where the n raw features are projected by P onto a smaller feature bond. Time is represented by a Fourier basis projected onto a time bond, and the resulting time-dependent kernels are applied through a full-lattice circular convolution evaluated with the fast Fourier transform, allowing every coefficient site to interact with every other site. Kernel coefficients are tied across the orbits of the square-lattice D_4 symmetry group, so with $*$ denoting periodic convolution, the velocity is

$$v(x, t; y) = G_\theta(t) * F(x) + H_\theta(t) * F(y). \tag{46}$$

The state and log-density equations are integrated together with a fixed-step fourth-order Runge–Kutta scheme (RK4), and because all kernel banks are initialized to zero, every transformation is initially the identity map. A stage therefore begins from the embedded parent augmented by the conditional free-theory detail source.

In Eq. 46, the condition enters additively. The conditional term has no dependence on x and therefore disappears under differentiation, leaving a divergence that contracts the center coefficients of the state kernel with the feature derivatives:

$$\nabla_x \cdot v(x, t; y) = \sum_{\text{sites}} \sum_p \left[G_\theta(t)\right]_{p,\text{center}} F_p'(x). \tag{47}$$

The divergence is therefore algebraic, and at a fixed state x, the condition y contributes nothing explicitly to Eq. 47; it does, however, change the trajectory x_t through the additive term, so the integrated divergence and endpoint Jacobian may still depend on y indirectly.

4.5 Density: Reconstruction and Composition of the Stage Log Density

The physical fine field is reconstructed only after both coefficient transports:

$$X_{2L} = W_{\mathrm{IRUV}}^{-1}(c_1, d_1). \tag{48}$$

The relevant Jacobians are the fixed-conditioning partial derivatives

$$J_{\mathrm{UV}} = \left.\frac{\partial d_1}{\partial d_0}\right|_{c_0}, \qquad J_{\mathrm{IR}} = \left.\frac{\partial c_1}{\partial c_0}\right|_{d_1}. \tag{49}$$

Because the UV and IR updates are sequential triangular maps with no cross term between them, their determinant contributions multiply, making Eq. 50 an exact change of variables from the parent field and detail source to the complete fine configuration space.

For the orthonormal Haar convention, the complete density after one stage is

$$\begin{aligned} \log q_{2L}(X_{2L}) = {} & \log q_L(X_L) + \log \pi_{\mathrm{free}}(d_0 \mid c_0) \\ & + \Delta \log q_{\mathrm{UV}} + \Delta \log q_{\mathrm{IR}} - L^2 \log 2. \end{aligned} \tag{50}$$

The final term is the fixed parent-to-IR Jacobian, while the full Haar transform introduces no additional contribution. Recursively applying Eq. 50 composes the base-model density with every accepted UV/IR stage, yielding the density from which importance weights and the ESS are computed. Both the state and log density are evaluated by numerical RK4 integration, whose integrated Jacobian differs slightly from the exact determinant of the discrete RK4 map, as discussed in Section 7.7.

4.6 Loss Function: Stagewise Reverse-KL Training with a Frozen Parent

The complete reverse-KL loss for one transition is

$$\mathcal{L}_{L\to 2L} = \frac{1}{(2L)^2}\mathbb{E}_{X_{2L}\sim q_{2L}}\left[S_{2L}(X_{2L}) + \log q_{2L}(X_{2L})\right]. \tag{51}$$

During training of a new stage, the parent sampler q_L is frozen, making $\log q_L(X_L)$ constant with respect to the new UV and IR parameters and therefore removable without changing their gradients. Define

$$\log q_{\text{stage}} = \log \pi_{\text{free}}(d_0 \mid c_0) + \Delta \log q_{\text{UV}} + \Delta \log q_{\text{IR}} - L^2 \log 2. \tag{52}$$

The resulting stagewise-equivalent loss is

$$\widetilde{\mathcal{L}}_{L\to 2L} = \frac{1}{(2L)^2}\mathbb{E}\left[S_{2L}(X_{2L}) + \log q_{\text{stage}}\right]. \tag{53}$$

A generated sample is passed through both coefficient transformations before the loss is evaluated, and both blocks are updated in one backward pass while previously accepted stages remain frozen. Consequently, a finer resolution can be added without backpropagation through, or reoptimization of, the existing hierarchy.

In every reported run, parents are drawn uniformly from a finite pool of frozen-model samples; Appendix F gives the more general law from which they may be drawn and identifies its endpoints.

4.7 Checkpoint Selection: Smoothed Training Loss

For every run, the checkpoint with the lowest training loss smoothed over three logged evaluations is accepted. Because no selection holdout is constructed and no reference Markov chain data are used, the procedure remains applicable when such data are unavailable.

5 Experimental Setup

5.1 Target and Architecture

The two-dimensional ϕ^4 target of Section 2.1 is used in all experiments at the near-critical couplings of Eq. 4.

The velocity field of Section 4.4 is used for every transformation, at every lattice size and in both the UV and IR roles, with its settings listed in Table 2. The table also gives the configuration keys from the released code, using separate names where the direct and lifting implementations differ. Architecture and optimizer values are fixed across the primary comparisons, and only the training seed is varied in Appendix I.1.

The velocity field contains one convolutional layer whose kernel spans the full lattice, so the velocity at one site may depend on the field values at all L^2 sites. This convolution is evaluated by fast Fourier transform, and all of its weights are initialized to zero.

5.2 Experimental Design and Training Protocol

Table 3 lists five sets of runs for which the backbone, solver grid and evaluation protocol are held fixed. Every lattice size and experimental arm uses an 80-step fixed-grid Runge–Kutta integrator, whose convergence, together with that of the single-scale training budgets, is checked in Appendix I.3.

A single-scale flow is compared with a layerwise lift at the same target size. Both are trained for 6000 updates at batch size 64, matching the update count and minibatch size but not the parameter count or total compute. Each ablation arm is paired with the layerwise lift at the same target size and on the same parent, so the pair differs only in the quantity varied by that ablation. Every lifting stage and ablation arm is appended to a checkpoint from Appendix H, where each parent was trained with the largest batch admitted by its lattice size.

One unconditional flow is trained at each of $L = 8$, 16, 32 and 64. At $L = 64$, a hardware-memory limit reduces the batch size to 32, and the update count is increased to 10000 to compensate partially; these runs are reported in Section 6.1.

At each of three transitions, one stage is appended to a frozen parent and trained with the physical action at the finer lattice size, after which the accepted stage is frozen; the results are reported in Section 6.2.

Table 2: Resolved architecture, solver and optimizer settings. The left column gives the configuration key used in the released code. A slash separates the direct-flow and lifting-stage keys where they differ.

configuration key	value	meaning
Velocity field		
`transportation_backbone`	`analytic_cnf`	velocity field of Eq. 46
`analytic_cnf_field_features`	64	per-site functions of the field value, one linear and 63 sines of trainable frequency
`analytic_cnf_field_bond`	32	channel width those 64 are projected to
`analytic_cnf_time_features`	25	Fourier basis functions of the flow time t
`analytic_cnf_time_bond`	24	channel width those 25 are projected to
`analytic_cnf_spatial_mode`	`dense_fft`	receptive field is the whole lattice
`analytic_cnf_d4_symmetric`	`True`	kernel weights tied as in Section 4.4
Solver		
`cnf_solver`	`rk4`	fixed-step fourth-order Runge-Kutta
`cnf_steps` / `uvir_cnf_steps`	80	direct flow / each lifting transport
Source and optimizer		
`init_type`	`free_theory`	source is a Gaussian free field
`init_free_m2` / `uvir_free_m2`	1.0	direct source / conditional detail source mass squared
`lr`	5×10^{-5}	AdamW learning rate
`grad_clip`	1.0	gradient-norm clipping threshold
`seed`	1234 / 4567–4570	base runs / primary lift and seed replicates

In the frozen-transport ablation, one transport is trained and the other is held at its identity initialization at all three transitions. Because the frozen transport remains at zero initialization, each arm can be regarded as a single-transport stage; the results are reported in Section 6.3.

In the joint-training ablation, the parent is unfrozen and the composed density is optimized with the finest action under warm and cold initializations. The warm arm uses the accepted coarse checkpoint, trained for 3000 updates at $L = 4$ and 6000 updates at $L = 8$, whereas the cold initialization receives only 100 updates; the results are reported in Section 6.4.

In the FiLM ablation, the additive conditioning of Eq. 46 is replaced by a multiplicative gain and tested under each parent schedule, with every variant compared against the additive arm under the same schedule in Section 6.5.

When a parent pool is used, it is generated once before training, held fixed, and sampled uniformly.

5.3 Evaluation Metrics and Training Reproducibility

Every reported number is obtained from $N = 10^4$ generated configurations, using sampling seed 482014 and generation batch size 64, without intermediate rejection, gating or resampling. Samples at every represented scale are retained from shared generative trajectories, and all metrics below are computed from them, so each intermediate scale is evaluated on the path used to produce the final configurations. Reference Markov chain ensembles are introduced only after training, with the ensemble used at each lattice size recorded in Appendix G.

Near the critical point of Section 2.1, structure is present at two distinct levels, either of which a generated ensemble can reproduce without the other. Global sector balance requires the two $\mathbb{Z}_2$ sectors of Eq. 2 to be populated in the correct proportion, whereas within-sector accuracy concerns fluctuations carried by long-wavelength modes whose correlation length is comparable to the lattice size at the chosen couplings. Finding both sectors is therefore sufficient for global balance, but the more demanding second property requires accurate configurations within each sector.

The following quantities are reported:

- ESS/N: the normalized effective sample size computed from the importance weights as $\mathrm{ESS}/N = (\sum_i w_i)^2/[N \sum_i w_i^2]$, which measures the efficiency of an importance-weighted estimator at the lattice size being evaluated.
- δ^2: the incremental log-weight-variance cost per created DOF,

$$\delta^2 = \frac{\sigma^2_{\log w}(\mathrm{out}) - \sigma^2_{\log w}(\mathrm{in})}{n}, \tag{54}$$

Table 3: Resolved training budgets, with rows within a block sharing every listed quantity except the one varied by that block. A dash in the pool column marks runs that draw parents live with the computational graph retained, while the single-scale flows at $L = 8$, 16 and 32 carry the budget of a lifting stage. The parent checkpoints to which every lift and ablation arm is appended are the larger-batch runs of Appendix H and are not listed here.

run	target L	updates	batch	parent pool
Single-scale flows				
flow $L = 8$	8	6000	64	—
flow $L = 16$	16	6000	64	—
flow $L = 32$	32	6000	64	—
flow $L = 64$	64	10000	32	—
Layerwise lifts				
lift $4 \to 8$	8	6000	64	131072
lift $8 \to 16$	16	6000	64	131072
lift $16 \to 32$	32	6000	64	131072
Ablation: frozen transport				
UV only, $4 \to 8$	8	6000	64	131072
IR only, $4 \to 8$	8	6000	64	131072
UV only, $8 \to 16$	16	6000	64	131072
IR only, $8 \to 16$	16	6000	64	131072
UV only, $16 \to 32$	32	6000	64	131072
IR only, $16 \to 32$	32	6000	64	131072
Ablation: joint training				
warm $4 \to 8$	8	6000	64	—
warm $8 \to 16$	16	6000	64	—
cold $4 \to 8$	8	6000	64	—
cold $8 \to 16$	16	6000	64	—
Ablation: FiLM conditioning				
layerwise $8 \to 16$	16	6000	64	131072
joint $8 \to 16$	16	6000	64	—

where n is the number of coordinates created by the model, and $\sigma^2_{\log w}(\mathrm{in})$ is the inherited log-weight variance; this equation is an exact accounting identity. A direct single-scale flow at lattice size L begins from the source p_0 of Section 2.2.2 and creates $n = L^2$ coordinates, giving the from-scratch reference cost $\sigma^2_{\log w}(L)/L^2$. A lift from L to $2L$ instead inherits a frozen parent and creates $n = 3L^2$ detail coefficients, giving the incremental cost $\delta^2 = [\sigma^2_{\log w}(2L) - \sigma^2_{\log w}(L)]/(3L^2)$, with the parent variance measured against its own physical action. Unlike either full variance, this difference is not itself a divergence: for paired parent and output draws, defining $R_{L\to 2L} = \log w_{2L} - \log w_L$ decomposes it into residual-variance and covariance terms. Appendix I.2 measures this decomposition and finds a covariance correction below 1.3% in every frozen-parent arm, supporting the use of δ^2 as an empirical diagnostic of the variance introduced by one lift. Values are reported only for direct flows and frozen-parent lifts, because a joint arm has no fixed inherited baseline to which a value could be assigned.

- $S = S_L(\phi)/L^2$: the mean action per site defined in Eq. 2, quoted as a relative difference from the MCMC reference in Appendix G; because it averages local terms over L^2 sites, it constrains one-site and nearest-neighbor statistics.
- $\langle m \rangle$: the magnetization defined in Eq. 6, reported through $|\langle m \rangle|/\langle |m| \rangle$, which is zero when the two sectors are balanced and approaches one when either is empty, thereby isolating global sector balance.
- $\langle |m| \rangle$: the mean absolute magnetization, quoted as a relative difference from the MCMC reference in Appendix G, which measures the average distance of the magnetization from zero. By contrast, χ measures the variance around that location, allowing a displaced magnetization to be distinguished from one with the correct location but the wrong spread.
- $\sqrt{\chi}/L$: the width of the magnetization inside one sector, formed from the sector-width susceptibility $\chi = L^2\left(\langle m^2 \rangle - \langle |m| \rangle^2\right)$, quoted as a relative difference from the MCMC reference in Appendix G. Subtracting $\langle |m| \rangle^2$ removes the sector location, so χ/L^2 is the variance of m around that location and $\sqrt{\chi}/L$ is the

standard deviation of $|m|$. This width is reported instead of χ because it directly measures the within-sector spread visible in the magnetization histogram; for small errors, its relative error is half that of χ, while for larger errors it is less than half.

- U_4: the Binder cumulant defined in Eq. 7, quoted as a relative difference from the MCMC reference in Appendix G. It equals $2/3$ when the magnetization is concentrated on two values without spread and decreases toward zero as the distribution broadens, so U_4 reports the shape of the within-sector spread while $\sqrt{\chi}/L$ reports its size.
- Classifier two-sample tests (C2STs): the held-out accuracy of logistic regressions trained to separate generated configurations from MCMC references. The original observable diagnostic, $\mathrm{C2ST}_{\mathrm{obs}}$, uses S, m and $|m|$ and therefore tests only the joint distribution of these three reported scalar observables.

  A second, post-analysis sensitivity diagnostic, $\mathrm{C2ST}_{\mathrm{MS}}$, uses a fixed fifteen-dimensional feature map whose local part contains m, $|m|$, m^2, m^4, the spatial means of ϕ^2 and ϕ^4, and the nearest-neighbor gradient energy per site. Its spectral part contains the logarithm of the mean Fourier power in the three integer momentum shells $|\boldsymbol{n}|^2 = 1, 2, 4$ and in five radial momentum bands, whose boundaries are 0, $1/8$, $1/4$, $3/8$ and $1/2$ cycles per lattice spacing, followed by a final band above $1/2$. Spatial averaging and Fourier magnitudes make these features translation invariant, radial averaging respects the square-lattice D_4 symmetry, and signed m retains sensitivity to a $\mathbb{Z}_2$ sector imbalance.

  Both feature maps use standardization fitted on the training split and the same classifier protocol, with each result averaged over five draws of 5000 configurations per class, a test fraction of 0.3, and every generated configuration entering unweighted. An accuracy near 0.5 means only that the classifier does not distinguish the ensembles in the selected features.

Training reproducibility is measured on four $4 \to 8$ layerwise lifts that share the configuration, budget, parent checkpoint and evaluation seed but vary the random seed used for stage training. As reported in Appendix I.1, the resulting relative spreads are 6.3% in ESS/N and 6.7% in δ^2, establishing the empirical resolution of this comparison. These replicates provide a reference for other transitions and architectures rather than an uncertainty model for them.

Cost is reported through peak allocated memory at the batch size of each run, and the training budget through the product of updates and batch size; elapsed times are omitted because the runs were dispatched to the heterogeneous hardware pool described in Appendix J.

6 Results

6.1 Direct Single-Scale Flows across Lattice Sizes

One unconditional flow is trained directly with the physical action at each lattice size in Table 4. At $L = 8$, 16 and 32, the per-DOF cost δ^2 of Eq. 54 is 3.46×10^{-4}, 4.62×10^{-4} and 4.34×10^{-4}, respectively, while ESS/N decreases from 0.977 to 0.590.

Width errors of $+2.3\%$, $+7.6\%$ and $+4.4\%$ accompany Binder errors of -2.2%, -3.3% and -2.4%, while the action error remains within half a percent and the mean absolute magnetization is underestimated by 1.8%, 3.4% and 5.4%. The corresponding $\mathrm{C2ST}_{\mathrm{obs}}$ values are 0.508, 0.512 and 0.540, while $\mathrm{C2ST}_{\mathrm{MS}}$ gives 0.506, 0.526 and 0.565, so both begin near the chance value of 0.5 and resolve increasing separation by $L = 32$. Across all four runs, peak memory is linear in the number of sites at 5×10^{-4} GiB per sample and site.

The absolute mean magnetization remains below 0.010 here and in every other reported run, whereas populating only one $\mathbb{Z}_2$ sector would give $\langle m \rangle = \pm\langle |m| \rangle$. Since $\langle |m| \rangle$ lies between 0.44 and 0.68 in the reference ensembles of Table 14, the measured imbalance is below three percent of the single-sector value in every case, confirming that global sector balance is retained throughout.

Table 4: Single-scale reference in which one unconditional flow is trained directly with the physical action at each lattice size. Training budgets are given in Table 3.

L	ESS/N	δ^2	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST}_{\mathrm{obs}}/\mathrm{C2ST}_{\mathrm{MS}}$	peak memory
8	0.977	3.46×10^{-4}	-0.5%	$+0.005$	-1.8%	$+2.3\%$	-2.2%	0.508/0.506	2.08 GiB
16	0.899	4.62×10^{-4}	$+0.2\%$	$+0.007$	-3.4%	$+7.6\%$	-3.3%	0.512/0.526	8.01 GiB
32	0.590	4.34×10^{-4}	$+0.1\%$	$+0.000$	-5.4%	$+4.4\%$	-2.4%	0.540/0.565	31.51 GiB
64	0.019	7.39×10^{-3}	-1.8%	$+0.002$	$+29.5\%$	-64.0%	$+18.0\%$	0.719/0.776	64.33 GiB

Over the same two doublings, ESS/N decreases by a factor of 1.1 from $L = 8$ to $L = 16$ and by a further factor of 1.5 from $L = 16$ to $L = 32$. This decrease is consistent with the volume factor in Eq. 18 acting on a relatively stable δ^2 between 3.5×10^{-4} and 4.6×10^{-4}. At the same three sizes, however, the larger-batch runs in Appendix H reach $\delta^2 \approx 2.5 \times 10^{-4}$, indicating that the per-site variance rate depends on training exposure.

At $L = 64$, the model collapses onto a small part of the target, a known failure mode of reverse-KL training that is particularly severe at high dimension [14, 12, 13]. Its δ^2 is 7.39×10^{-3}, or 17 times the value at $L = 32$, while ESS/N falls to 0.019.

The generated Binder cumulant is 0.654, compared with the reference value of 0.554; because a magnetization concentrated on only two values would saturate at $2/3$, the generated model is close to this limit. Its width is 0.060 against a reference value of 0.168, the standard deviation of its action per site is 14% below the reference, and its mean absolute magnetization is overestimated by 29.5%. Here $\mathrm{C2ST}_{\mathrm{obs}} = 0.719$ and $\mathrm{C2ST}_{\mathrm{MS}} = 0.776$, well above their respective ranges of 0.508–0.540 and 0.506–0.565 at the three smaller sizes, although the mean magnetization remains 0.002 and shows that both $\mathbb{Z}_2$ sectors are still produced.

Every comparison that uses the single-scale flow as a reference is therefore restricted to $L = 8$ through $L = 32$, over which δ^2 remains between 3.5×10^{-4} and 4.6×10^{-4}.

6.2 Layerwise Lifting Flows across Lattice Sizes

Table 5 appends one lifting stage to a frozen base at each of three transitions and compares it with the single flow trained directly at the same target size.

Table 5: Layerwise lifting stages compared with the single-scale flow at the same target lattice. The ratio is the stage value of δ^2 divided by the value reached by the single flow in Table 4 at the same target size.

transition	ESS/N	$\delta^2/10^{-2}$	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST}_{\mathrm{obs}}/\mathrm{C2ST}_{\mathrm{MS}}$	peak memory	ratio
$4 \to 8$	0.460	1.51	$+1.9\%$	-0.006	-13.2%	$+3.5\%$	-11.8%	0.568/0.630	3.65 GiB	44
$8 \to 16$	0.050	1.45	$+1.4\%$	-0.001	-9.1%	$+1.7\%$	-4.5%	0.550/0.716	13.38 GiB	31
$16 \to 32$	0.0029	1.48	$+1.4\%$	-0.002	-8.4%	-2.4%	-1.6%	0.551/0.871	50.23 GiB	34

The composed ESS/N decreases from 0.460 to 0.0029 over the three transitions. In units of 10^{-2}, δ^2 takes the values 1.51, 1.45 and 1.48, with a spread of 4.1% about their mean. These values are 44, 31 and 34 times those of the single flow at the same target size, making the incremental cost of a lifting stage per generated DOF, on average, about 36 times the from-scratch cost of a single flow per site. This factor does not decrease as the lattice is enlarged under the matched update and batch schedule.

Across the three transitions, the action error remains between $+1.4\%$ and $+1.9\%$, while the width errors are $+3.5\%$, $+1.7\%$ and -2.4%, and the Binder error approaches zero from -11.8% to -1.6%. The mean absolute magnetization is underestimated by 13.2%, 9.1% and 8.4%, with corresponding $\mathrm{C2ST}_{\mathrm{obs}}$ values of 0.568, 0.550 and 0.551, showing that agreement in the three original observables improves once and then plateaus, whereas $\mathrm{C2ST}_{\mathrm{MS}}$ increases from 0.630 to 0.716 and 0.871. Thus, the contraction of the magnetization and Binder errors does not extend to every measured discrepancy, and at the same target sizes the single flow yields lower $\mathrm{C2ST}_{\mathrm{MS}}$ values of 0.506, 0.526 and 0.565. The mean magnetizations of -0.006, -0.001 and -0.002 confirm that global sector balance is retained at every transition and that the reported errors are within-sector discrepancies.

Peak memory for the stages is 3.65, 13.38 and 50.23 GiB. The corresponding single-flow values in Table 4 are 2.08, 8.01 and 31.51 GiB, so the stages use 1.75, 1.67 and 1.59 times as much memory. Their parameter counts are 79150, 189742 and 558382, compared with 14231, 37271 and 120215 for the single flows, corresponding to factors of 5.6, 5.1 and 4.6. Since neither ratio decreases with lattice size, the stage remains larger and more expensive at every transition.

6.3 Ablation: Relative Importance of the UV and IR Transports

The two frozen-transport arms are reported in Table 6 and compared at each transition with the layerwise lift of Section 6.2, using the same parent and training schedule. The UV-only arm trains the transformation that generates the new coefficients while holding coarse relaxation at the identity, whereas the IR-only arm uses the converse arrangement.

When the IR transport is held at the identity, δ^2 reaches 29.46×10^{-2}, 29.35×10^{-2} and 30.14×10^{-2}. These values are 19.5, 20.2 and 20.3 times those of the full stage. When the UV transport is held at the identity, δ^2 reaches 2.24×10^{-2}, 2.16×10^{-2} and 2.26×10^{-2}, corresponding to factors of 1.48, 1.48 and 1.53. Both factors remain stable across the three transitions, with the UV-only and IR-only values spanning 2.7% and 4.8% about their respective means.

Table 6: Frozen-transport arms against the layerwise lift at the same transition, with the ratio column giving δ^2 divided by the value reached by the full stage at that transition.

transition	trained	ESS/N	$\delta^2/10^{-2}$	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST}_{\mathrm{obs}}/\mathrm{C2ST}_{\mathrm{MS}}$	ratio
$4 \to 8$	UV and IR	0.460	1.51	$+1.9\%$	-0.006	-13.2%	$+3.5\%$	-11.8%	0.568/0.630	1.00
	UV only	0.053	29.46	$+22.6\%$	-0.009	$+10.8\%$	$+33.9\%$	-13.7%	0.667/0.762	19.5
	IR only	0.195	2.24	-13.5%	-0.007	-2.2%	$+6.3\%$	-3.6%	0.627/0.673	1.48
$8 \to 16$	UV and IR	0.050	1.45	$+1.4\%$	-0.001	-9.1%	$+1.7\%$	-4.5%	0.550/0.716	1.00
	UV only	0.00066	29.35	$+24.1\%$	-0.001	$+11.2\%$	$+30.1\%$	-7.7%	0.823/0.920	20.2
	IR only	0.013	2.16	-14.2%	-0.002	$+6.9\%$	$+1.8\%$	$+3.2\%$	0.736/0.811	1.48
$16 \to 32$	UV and IR	0.0029	1.48	$+1.4\%$	-0.002	-8.4%	-2.4%	-1.6%	0.551/0.871	1.00
	UV only	0.00011	30.14	$+22.1\%$	-0.002	$+14.9\%$	$+24.5\%$	-2.6%	0.957/0.993	20.3
	IR only	0.00071	2.26	-14.8%	-0.002	$+13.8\%$	$+7.0\%$	$+3.5\%$	0.901/0.965	1.53

For the UV-only arm, the composed ESS/N decreases to 0.053, 0.00066 and 0.00011. Its width errors are $+33.9\%$, $+30.1\%$ and $+24.5\%$, whereas for the IR-only arm, ESS/N is 0.195, 0.013 and 0.00071, with width errors of $+6.3\%$, $+1.8\%$ and $+7.0\%$. At the largest transition, these ESS values correspond to approximately one and seven effective samples among the 10^4 evaluated configurations, so the resulting estimates may be biased.

The action errors have opposite signs in the two frozen arms at every transition, ranging from $+22.1\%$ to $+24.1\%$ when the IR transport is held at the identity and from -13.5% to -14.8% when the UV transport is held there, compared with $+1.4\%$ to $+1.9\%$ for the full stage. Width errors are positive in both frozen arms at every transition and reach between $+24.5\%$ and $+33.9\%$ when the IR transport is held at the identity.

For the full stage, the Binder error ranges from -11.8% to -1.6%. The range is -13.7% to -2.6% when the IR transport is held at the identity and -3.6% to $+3.5\%$ when the UV transport is held there. With the IR transport frozen, the mean absolute magnetization is overestimated by 10.8%, 11.2% and 14.9%, whereas with the UV transport frozen its error ranges from -2.2% to $+13.8\%$, compared with -13.2% to -8.4% for the full stage.

When the IR transport is held at the identity, $\mathrm{C2ST}_{\mathrm{MS}}$ gives 0.762, 0.920 and 0.993. When the UV transport is held there, values of 0.673, 0.811 and 0.965 are obtained, compared with 0.630, 0.716 and 0.871 for the full stage. The same ordering is obtained with $\mathrm{C2ST}_{\mathrm{obs}}$, whose values are listed first in Table 6. Across all nine arms, the mean magnetization remains between -0.009 and -0.001, confirming that global sector balance is retained under both ablations and that the reported errors are within-sector discrepancies.

6.4 Ablation: Layerwise versus Joint Training

In Table 7, the layerwise lift of Section 6.2 is compared with two joint arms at the same transitions while the architecture, update count and batch schedule are held fixed. Each joint arm unfreezes every level and optimizes the composed log density against the finest action alone; the warm arm begins from the accepted coarse checkpoint, trained for 3000 updates at $L = 4$ and 6000 updates at $L = 8$, whereas the coarse level of the cold arm receives only 100 updates. In a separate validation, this cold level reaches $\mathrm{ESS}/N = 0.006$ after those 100 updates.

Table 7: Layerwise and joint training at fixed architecture. The rows differ only in whether the parent is frozen and, for the joint arms, how it is initialized. A dash marks joint arms, for which δ^2 would subtract a parent variance measured against an action that the parent no longer approximates.

schedule	target L	ESS/N	$\delta^2/10^{-2}$	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST}_{\mathrm{obs}}/\mathrm{C2ST}_{\mathrm{MS}}$
layerwise	8	0.460	1.51	$+1.9\%$	-0.006	-13.2%	$+3.5\%$	-11.8%	0.568/0.630
joint, warm	8	0.518	—	-0.7%	-0.008	-0.2%	$+0.1\%$	-0.4%	0.508/0.611
joint, cold	8	0.424	—	$+0.9\%$	-0.007	-11.3%	-2.2%	-5.1%	0.559/0.611
layerwise	16	0.050	1.45	$+1.4\%$	-0.001	-9.1%	$+1.7\%$	-4.5%	0.550/0.716
joint, warm	16	0.115	—	-0.3%	-0.001	-2.6%	$+2.2\%$	-1.6%	0.522/0.699
joint, cold	16	0.094	—	$+0.1\%$	-0.001	-7.2%	$+1.0\%$	-2.9%	0.545/0.708

At the coarser transition, ESS/N is 0.460, 0.518 and 0.424 for the layerwise, warm joint and cold joint schedules, respectively. At the finer transition, the corresponding values are 0.050, 0.115 and 0.094, again making the warm joint

arm the highest of the three. Its improvements of 0.058 and 0.065 over the layerwise arm account for 11% and 7.6% of the gap to the single-scale flow in Table 4.

At the coarser transition, the width error changes from $+3.5\%$ under the layerwise schedule to $+0.1\%$ under the warm joint schedule, while the Binder error changes from -11.8% to -0.4%; in the cold joint arm, these errors are -2.2% and -5.1%, respectively. At the finer transition, the layerwise and warm joint width errors are $+1.7\%$ and $+2.2\%$, while the Binder error changes from -4.5% to -1.6%.

The action error remains within one percent in every joint arm, compared with $+1.9\%$ and $+1.4\%$ in the layerwise arms. In the layerwise, warm and cold ordering, the mean absolute magnetization errors are -13.2%, -0.2% and -11.3% at the coarser transition and -9.1%, -2.6% and -7.2% at the finer transition. The corresponding $\mathrm{C2ST}_{\mathrm{obs}}$ values are 0.568, 0.508 and 0.559 at the coarser transition, and 0.550, 0.522 and 0.545 at the finer transition. For the same layerwise, warm and cold ordering, $\mathrm{C2ST}_{\mathrm{MS}}$ gives 0.630, 0.611 and 0.611 at $L = 8$, and 0.716, 0.699 and 0.708 at $L = 16$, showing that joint training improves the selected magnetization observables much more strongly than it improves multiscale distinguishability. Since the mean magnetization ranges from -0.006 to -0.008 at the coarser transition and is -0.001 at the finer transition, the residual discrepancies are within-sector errors rather than an imbalance between the global sectors.

6.5 Ablation: Additive versus FiLM Conditioning

Because the condition enters Eq. 46 additively, it contributes no explicit term to the instantaneous divergence at a fixed state, although it can still affect the integrated Jacobian indirectly by changing the trajectory. Appendix E.2 shows that the target conditional width varies by roughly one fifth with local detail energy, which motivates testing whether a direct condition-dependent multiplicative pathway, added to the dependence already induced through the trajectory, reduces the incremental lift cost.

The ablation replaces the additive form by a gain constructed from the condition and time, which multiplies the state-dependent term as in feature-wise linear modulation (FiLM) [45]. Because this gain does not depend on the transported field, the divergence remains available in closed form, as detailed in Appendix C. Both training schedules are evaluated at the $8 \to 16$ transition with identical budgets, seeds and accepted steps.

Table 8: Effect of replacing additive conditioning with a multiplicative gain at the $8 \to 16$ transition. A dash marks the joint arms. For those arms, the definition of δ^2 would subtract a parent variance measured against an action that the parent no longer approximates.

schedule	conditioning	ESS/N	$\delta^2/10^{-2}$	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST}_{\mathrm{obs}}/\mathrm{C2ST}_{\mathrm{MS}}$
layerwise	additive	0.050	1.45	$+1.4\%$	-0.001	-9.1%	$+1.7\%$	-4.5%	0.550/0.716
	multiplicative	0.017	1.47	$+1.8\%$	-0.001	-10.7%	$+1.5\%$	-5.5%	0.560/0.719
joint	additive	0.115	—	-0.3%	-0.001	-2.6%	$+2.2\%$	-1.6%	0.522/0.699
	multiplicative	0.096	—	-0.2%	-0.001	-4.4%	$+0.0\%$	-1.7%	0.530/0.703

With the multiplicative gain, ESS/N changes from 0.050 to 0.017 under the layerwise schedule and from 0.115 to 0.096 under the joint schedule, corresponding to decreases of 66% and 16%, respectively.

Values of 1.45×10^{-2} and 1.47×10^{-2} are obtained for δ^2 in the two layerwise arms, which agree within two percent and both leave the stage a factor of 32 above the 4.62×10^{-4} reached by a single flow under the same update and batch schedule at this size.

Under the joint schedule, the width error changes from $+2.2\%$ to $+0.0\%$, whereas under the layerwise schedule it changes from $+1.7\%$ to $+1.5\%$. The layerwise Binder error changes from -4.5% to -5.5%, while the two joint arms give -1.6% and -1.7%; similarly, the mean absolute magnetization error changes from -9.1% to -10.7% under the layerwise schedule and from -2.6% to -4.4% under the joint schedule. The corresponding $\mathrm{C2ST}_{\mathrm{obs}}$ changes are 0.550 to 0.560 and 0.522 to 0.530, while the smaller $\mathrm{C2ST}_{\mathrm{MS}}$ changes are 0.716 to 0.719 and 0.699 to 0.703. The action error remains within two percent in all four arms, and the mean magnetization is -0.001 in each case, so the form of conditioning does not affect global sector balance.

7 Discussion

7.1 Separating Incremental Cost from Lattice Volume

Equation 54 expresses the log-weight variance as $\sigma^2_{\log w} = n\delta^2 + \sigma^2_{\log w}(\mathrm{in})$, where n is the number of coordinates created by the model and the second term is the inherited variance, thereby separating the measured increase of a frozen-parent lift into a coordinate count and an incremental rate. Appendix I.2 shows that this rate agrees within 1.3% with the paired residual variance per new coefficient, while the log-normal approximation of Eq. 18 predicts that an approximately fixed rate produces an exponential decrease in ESS/N as more coefficients are created. This accounting is an empirical cost decomposition for the frozen-parent protocol rather than a divergence decomposition; the full output variance characterizes the endpoint sampler.

As reported in Table 4, a single flow gives δ^2 values of 3.46×10^{-4}, 4.62×10^{-4} and 4.34×10^{-4} at $L = 8$, 16 and 32. Across this range, ESS/N decreases from 0.977 to 0.590, while the per-site cost changes by one third and the site count grows sixteenfold, indicating that most of the efficiency decrease is associated with increased volume rather than an order-of-magnitude change in the per-site variance rate. At $L = 64$, by contrast, δ^2 reaches 7.39×10^{-3}, seventeen times the $L = 32$ value, and Section 6.1 identifies this isolated degradation as mode collapse.

As reported in Table 5, the lifting stages give δ^2 values of 1.51×10^{-2}, 1.45×10^{-2} and 1.48×10^{-2}, whose 4.1% spread lies within the 6.7% training-seed spread and therefore makes the per-coefficient cost stable at the precision of this experiment. Over the same transitions, the number of created coefficients grows from 48 to 768, while ESS/N decreases by a factor of 160, from 0.460 to 0.0029, consistently with a nearly fixed incremental rate applied to more coefficients. Relative to the from-scratch single-flow rate at the same target, the corresponding ratios of 44, 31 and 34 show no decrease with lattice size.

A second limitation of ESS/N appears at the largest transition. At $16 \to 32$, the composed variance is 11.46. Equation 18 then gives 1.1×10^{-5}, whereas 10^4 configurations yield 2.9×10^{-3}. At the same sample count, simulated log-normal weights with this variance yield a median of 2.0×10^{-3} despite a much smaller log-normal large-sample estimate. The reported value is therefore limited by sample size and should be regarded as an upper bound on the model's efficiency, whereas the per-coefficient cost is less affected because it is averaged over all 10^4 configurations rather than being carried by the largest few weights.

Two operational rates are therefore compared in Figure 3(a): the total variance per site for a from-scratch direct flow and the incremental variance per newly created coefficient for a frozen-parent lift. Although their accounting baselines differ by design, their ratio measures how much variance the tested lift adds per new coefficient relative to learning a site directly, while the full ESS/N remains the endpoint efficiency measure.

7.2 Observable Agreement and Multiscale Distinguishability

Global sector balance is retained in every reported arm: across all twenty-six arms, the absolute mean magnetization remains below 0.010, compared with a mean absolute magnetization between 0.44 and 0.76 in the references of Appendix G. The sector imbalance is therefore below three percent of the value obtained when only one sector is populated, so the failures reported below concern distributions within the sectors rather than the loss of a global sector.

The two C2ST feature maps answer different questions. The original $\mathrm{C2ST}_{\mathrm{obs}}$ tests separability in the three scalars already reported in the tables, whereas the sensitivity diagnostic $\mathrm{C2ST}_{\mathrm{MS}}$ additionally probes local field moments and Fourier power across scales. Neither is identified with full-distribution accuracy, and their difference instead measures sensitivity to the chosen projection.

As reported in Table 4, the single-flow $\mathrm{C2ST}_{\mathrm{obs}}$ values are 0.508, 0.512, 0.540 and 0.719 from $L = 8$ to $L = 64$. The corresponding $\mathrm{C2ST}_{\mathrm{MS}}$ values are 0.506, 0.526, 0.565 and 0.776. Both diagnostics are near chance at $L = 8$ and increase with size; from $L = 16$ onward, the broader feature map resolves the larger separation, so distinguishability and efficiency worsen in the same direction for this construction.

Up to $L = 32$, the shape errors remain small and keep the same sign, with the mean absolute magnetization underestimated by 1.8%, 3.4% and 5.4% and the width overestimated by 2.3%, 7.6% and 4.4%. The generated magnetization therefore lies slightly inside the reference with a slightly larger spread, as shown at $L = 32$ in Figure 2(a). Over the same range, the Binder error lies between -2.2% and -3.3%, consistently with the sign of the width error.

At $L = 64$, both errors reverse sign and grow by an order of magnitude, reaching $+29.5\%$ for the mean absolute magnetization and -64.0% for the width. The generated ensemble is therefore displaced outward and is much narrower than the reference, as shown in Figure 2(c). The Binder error also reverses sign and reaches $+18.0\%$, while $\mathrm{C2ST}_{\mathrm{obs}} = 0.719$ and $\mathrm{C2ST}_{\mathrm{MS}} = 0.776$, so both feature maps resolve the failure. This collapse establishes a limit

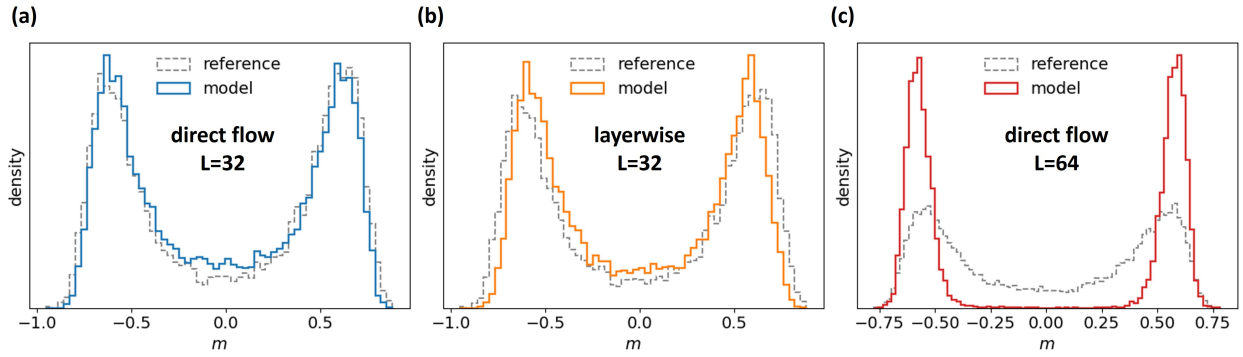


Figure 2: Magnetization densities generated by the direct and layerwise constructions and compared with the references. Panels (a) and (b) show the direct flow and layerwise lift, respectively, at $L = 32$; panel (c) shows the failed direct flow at $L = 64$. Histograms of m are formed from the 10^4 configurations described in Section 5.3 and are normalized to unit area. At $L = 32$, both constructions reproduce the two-sector shape and overlap substantially with the reference, although their density-based and multiscale diagnostics differ sharply. At $L = 64$, the direct-flow distribution is narrower and displaced outward.

for the single-scale reference at the tested batch size but not its behavior under a larger batch, and all subsequent comparisons are therefore restricted to $L = 8$ through $L = 32$.

As reported in Table 5, the lifting-stage $\mathrm{C2ST}_{\mathrm{obs}}$ values are 0.568, 0.550 and 0.551. The mean absolute magnetization error contracts from -13.2% to -8.4%, and the Binder error contracts from -11.8% to -1.6%. Agreement in this observable projection therefore improves after the first transition and then plateaus, while the width error crosses zero from $+1.7\%$ to -2.4%, changing the peaks from slightly broad to slightly narrow. The action error remains between $+1.4\%$ and $+1.9\%$. At $L = 32$, the substantial histogram overlap is visible in Figure 2(b): the direct and layerwise values of $\mathrm{C2ST}_{\mathrm{obs}}$ are similarly 0.540 and 0.551, but their ESS/N values are 0.590 and 0.0029, and their $\mathrm{C2ST}_{\mathrm{MS}}$ values are 0.565 and 0.871, respectively.

The multiscale feature map gives the opposite trend. Its accuracies increase from 0.630 to 0.716 and 0.871, compared with a training-seed span of 0.011 at the first transition, so smaller Binder and magnetization errors coexist with increasingly distinguishable local and spectral statistics, as shown for both projections in panel (b) of Figure 3.

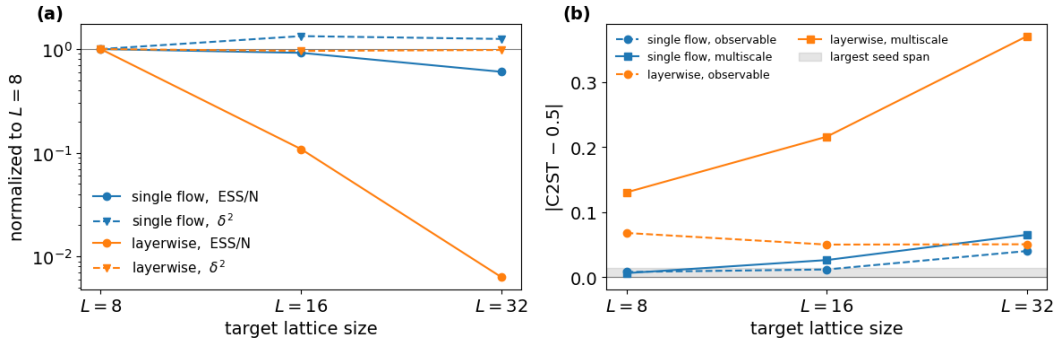


Figure 3: Efficiency and feature-level distinguishability as functions of target lattice size. Panel (a) shows ESS/N and δ^2, each divided by its value at $L = 8$, on a logarithmic axis. Panel (b) shows the C2ST excess above 0.5. Dashed lines use the observable map and solid lines the multiscale map. The shaded band marks the larger of their training-seed spans, 0.014, from Appendix I.1. For both constructions, δ^2 remains within one third of its initial value while ESS/N decreases. The observable projection hides most of the layerwise trend; the multiscale projection becomes progressively more distinguishable.

The apparent trend therefore depends on the feature map. Efficiency decreases from 0.977 to 0.590 for the single flow and from 0.460 to 0.0029 for the composed sampler, while both constructions become easier to distinguish in the multiscale map, with a much steeper increase for the composed sampler. At the same time, the composed sampler's low-order magnetization and Binder errors contract, so its best reproduction of these projections occurs where its

importance weights and broader feature agreement are least usable. This shall not be interpreted as a contradiction because agreement on a finite set of observables does not control the full density ratio.

7.3 Why Some Low-Order Errors Contract with Lattice Size

As reported in Section 7.2, several low-order errors of the composed sampler contract as the lattice is enlarged, as does one target-side component of the required stage correction: the rescaling-invariant Binder witness ΔU_4 in Table 1 decreases from 0.0642 to 0.0149. Meanwhile, the measured log-weight cost δ^2 remains stable within 4.1%, so the two trends reject the hypothesis that this Binder-shape mismatch quantitatively predicts the stage cost.

The required and remaining corrections can be compared directly because both are differences of Binder cumulants: the required values are 0.0642, 0.0365 and 0.0149, while absolute residuals of 0.0609, 0.0247 and 0.0090 remain after the stage. The surviving fraction therefore decreases from 0.95 to 0.60, with the required correction decreasing by a factor of 4.3 and the residual by a factor of 6.8.

This agreement is largely confined to the Binder cumulant. Across the same transitions, the mean absolute magnetization error decreases by a factor of 1.56, the width error by 1.45, and the action error by 1.37. The $\mathrm{C2ST}_{\mathrm{obs}}$ excess above 0.5 decreases by a factor of 1.34, whereas the $\mathrm{C2ST}_{\mathrm{MS}}$ excess grows by a factor of 2.84. These trends are compared with factors of 4.31 for ΔU_4 and 7.47 for the Binder error, so the closest agreement is obtained in the same observable from which ΔU_4 is constructed. The witness can therefore help organize the Binder residual but not the broader feature discrepancy.

The parent $\mathrm{C2ST}_{\mathrm{obs}}$ values are 0.502, 0.503 and 0.511, and the corresponding $\mathrm{C2ST}_{\mathrm{MS}}$ values are 0.504, 0.498 and 0.519. The parent Binder errors are -0.14%, -1.53% and -0.18%, an order of magnitude smaller than the stage residuals and nonmonotone, so the observed contraction is not inherited from a parent that improves with size. Absolute residuals also contract together with relative residuals; for example, the mean absolute magnetization residual decreases from 0.089 to 0.044 in field units, which excludes a merely shrinking reference scale as the explanation.

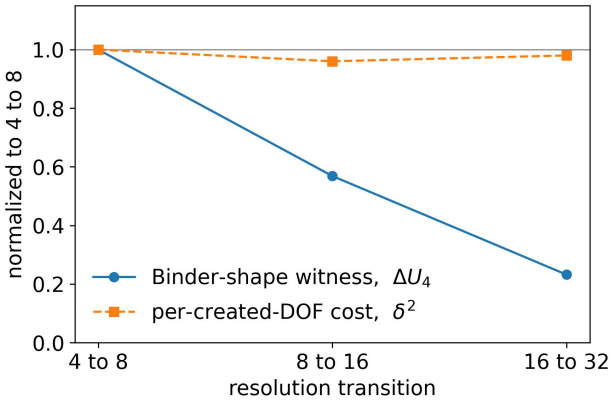


Figure 4: Binder-shape mismatch and measured stage cost at each transition. The scalar witness ΔU_4 from Table 1 and δ^2 from Table 5 are each divided by their value at $4 \to 8$. Both can therefore be shown on one axis. This Binder component decreases to 0.23 of its initial value, while the cost remains within 4% of its initial value. As defined in Section 2.4, ΔU_4 is not a distance between the coarse distributions.

The same measurements give the opposite conclusion for cost: a factor-4.31 decrease in ΔU_4 is accompanied by a change of only 4.1% in δ^2, as shown in Figure 4, while the amplitude ratios r_σ and r_{m^2} in Table 1 remain near 0.85 and 0.75. Because the stage must also generate UV details, correct the coarse variables conditional on those details, and approximate both transformations under finite optimization, a shrinking Binder-shape component need not reduce the full density-ratio cost.

The original expectation that a smaller ΔU_4 would make the lift cheaper is thus unsupported, although the witness remains useful for showing that a constant rescaling is insufficient and for explaining the contraction of a Binder-related residual. Its magnitude is therefore not a quantitative proxy for either δ^2 or full-distribution agreement.

7.4 The Roles of the Two Transports

As shown in Table 6 and Figure 5, the three arms are separated by δ^2 at every transition. When coarse relaxation is held at the identity, the per-coefficient cost is multiplied by 19.5, 20.2 and 20.3. When detail generation is held there, it is multiplied by 1.48, 1.48 and 1.53. Within each arm, the cost remains stable across the transitions, with spreads of 2.7% and 4.8% compared with 4.1% for the full stage. The decrease in ESS/N within each arm is again associated with the volume factor of Section 7.1, while δ^2 carries the separation between arms.

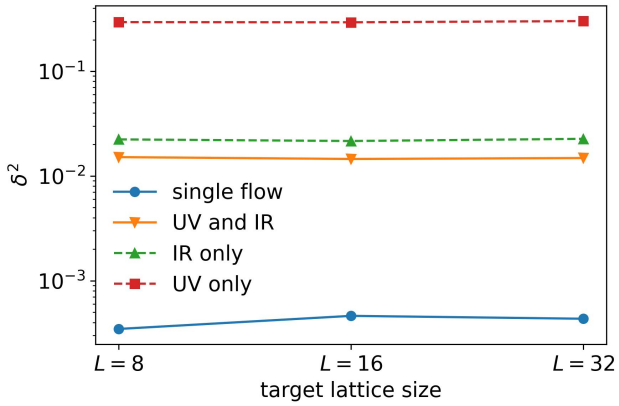


Figure 5: Per-coefficient cost of each arm at each transition, shown on a logarithmic axis. For the single flow, δ^2 lies between 3.46×10^{-4} and 4.62×10^{-4}. For the full stage, it lies between 1.45×10^{-2} and 1.51×10^{-2}. Values between 2.16×10^{-2} and 2.26×10^{-2} are obtained when the UV transport is held at the identity. When the IR transport is held there, values between 2.94×10^{-1} and 3.01×10^{-1} are obtained. Each series is constant across the three transitions within the 6.7% seed spread of Appendix I.1.

Across the nine arms, the mean magnetization remains between -0.009 and -0.001, so global sector balance survives both ablations. The action errors have opposite signs, ranging from $+22.1\%$ to $+24.1\%$ with coarse relaxation frozen and from -13.5% to -14.8% with detail generation frozen, while the full-stage errors lie between these ranges at $+1.4\%$ to $+1.9\%$. The two transformations therefore correct the action in opposite directions, and the trained pair lies between the single-transformation limits.

The transports are also separated by the magnetization features they control, as visualized in Figure 6. With relaxation frozen, the mean absolute magnetization errors range from $+10.8\%$ to $+14.9\%$. With generation frozen, they range from -2.2% to $+13.8\%$. For the full stage, the values lie below the reference at -13.2% to -8.4%. Thus, freezing either transformation generally places the sectors farther from zero than required, whereas the trained pair moves them past the reference in the opposite direction.

The width is controlled primarily by coarse relaxation. When relaxation is held at the identity, width errors of $+24.5\%$ to $+33.9\%$ are obtained, whereas holding generation there gives errors from $+1.8\%$ to $+7.0\%$.

The cost and classifier diagnostics give the same ordering of these three model classes. In $\mathrm{C2ST}_{\mathrm{MS}}$, holding relaxation at the identity gives 0.762, 0.920 and 0.993, while holding generation there gives 0.673, 0.811 and 0.965, compared with 0.630, 0.716 and 0.871 for the full stage. Both transformations therefore improve multiscale agreement, with the largest deficit again occurring without coarse relaxation, although all three arms become increasingly separable with size. The observable-map values in Table 6 preserve the same between-arm ordering but conceal most of the trend within the full stage.

With generation frozen, Binder errors of -3.6%, $+3.2\%$ and $+3.5\%$ are obtained. The full-stage values are -11.8%, -4.5% and -1.6%. At the coarsest transition, the arm with 1.48 times the cost therefore has the smaller Binder error, illustrating that a smaller error in one projected observable does not imply a smaller overall discrepancy.

Any modification aimed at reducing stage cost must therefore address coarse relaxation. Most of the contribution from the generating transport is already supplied by the conditional free-theory source of Section 4.2; nevertheless, both transformations are required, because freezing either one raises $\mathrm{C2ST}_{\mathrm{MS}}$ above the full-stage value at every transition and above 0.96 at the largest transition.

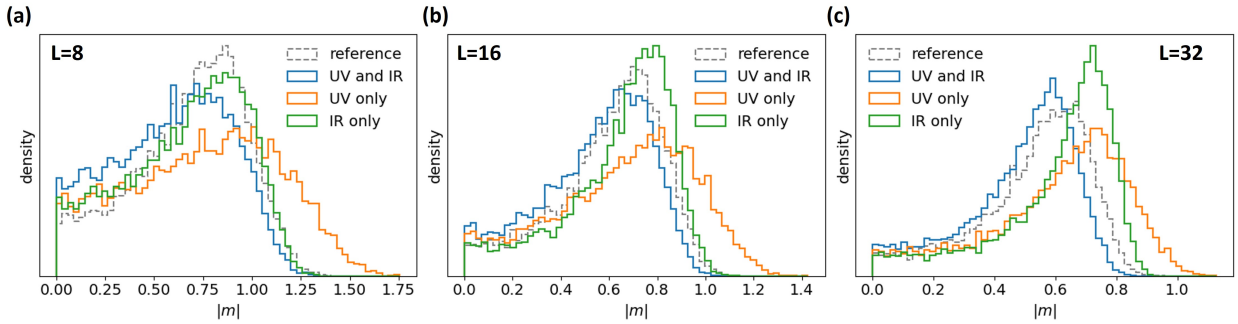


Figure 6: Magnetization densities of the full stage and the two frozen-transport arms. Results at $L = 8$, 16 and 32 are shown in panels (a), (b) and (c), respectively. Histograms of $|m|$ are formed from the 10^4 configurations described in Section 5.3 and are normalized to unit area. At every transition, the full-stage distribution lies inside the reference, whereas both frozen-arm distributions lie outside it. The broadest distribution is obtained when the IR transport is held at the identity.

7.5 Ablating the Training Schedule and the Conditioning

The construction fixes two design choices, the training schedule across levels and the form of conditioning in the velocity field, each of which is varied in turn while the architecture, budget and solver grid are held fixed.

The layerwise parameter setting is feasible for the jointly optimized reverse-KL problem, so the globally attainable joint objective can be no worse than the value at that setting. This argument neither orders the attained optimizers by ESS/N nor orders the finite direct and composed model classes by bijectivity alone.

The attained joint efficiency lies between the measured layerwise and direct values in Table 7. At the coarser transition, the layerwise and direct values are 0.460 and 0.977, while joint optimization reaches 0.518. At the finer transition, the corresponding values are 0.050, 0.899 and 0.115, so unfreezing the parent recovers 11% and 7.6% of the gap between the two constructions.

The selected observables are improved more strongly by the same change. At the coarser transition, $\mathrm{C2ST}_{\mathrm{obs}}$ changes from 0.568 to 0.508, the Binder error from -11.8% to -0.4%, the width error from $+3.5\%$ to $+0.1\%$, and the mean absolute magnetization error from -13.2% to -0.2%. At the finer transition, $\mathrm{C2ST}_{\mathrm{obs}}$ changes from 0.550 to 0.522, and the Binder error from -4.5% to -1.6%.

The multiscale changes are much smaller: the warm joint schedule moves $\mathrm{C2ST}_{\mathrm{MS}}$ from 0.630 to 0.611 at $L = 8$ and from 0.716 to 0.699 at $L = 16$, compared with direct-flow values of 0.506 and 0.526. Joint training therefore brings the magnetization projections close to their direct counterparts but leaves substantial multiscale separation, while its efficiency remains lower by factors of 1.9 at $L = 8$ and 7.8 at $L = 16$. These results are consistent with a contribution from the frozen inherited marginal to the low-order observable deficit, but they do not reduce the remaining discrepancy to that source alone.

Every arm in Section 6, together with the parent checkpoints, is placed on the two classifier projections in Figure 7. In the observable map, several composed arms overlap the direct flows despite much larger log-weight variance, producing the apparent decoupling associated with the three selected scalars. In the multiscale map, the ordering with log-weight variance is much tighter because the same arms move upward once local moments and spectral power are included, showing why chance-level performance of one feature-based classifier cannot guarantee a small density-ratio variance.

The cold arm separates the schedule comparison from the quality of the initial parent by giving its coarse level 100 updates, compared with the 3000 and 6000 updates used for the warm initializations. It reaches ESS/N values of 0.424 and 0.094, compared with 0.460 and 0.050 under the layerwise schedule, and at the coarser transition gives action, width and Binder errors of $+0.9\%$, -2.2% and -5.1%, compared with the corresponding layerwise errors of $+1.9\%$, $+3.5\%$ and -11.8%.

The three schedules span a frozen coarse level, a fully converged unfrozen level and a minimally trained unfrozen level; across these settings, fine-scale efficiency changes by one fifth at the coarser transition and by a factor of 2.3 at the finer transition, whereas whether the model is factorized at all produces larger factors of 2.1 and 18, respectively.

No change in the conditioning ablation is resolved beyond the seed spread. As reported in Table 8, the two layerwise values of δ^2 are 1.45×10^{-2} and 1.47×10^{-2}, while the largest observable change under either schedule is 1.61

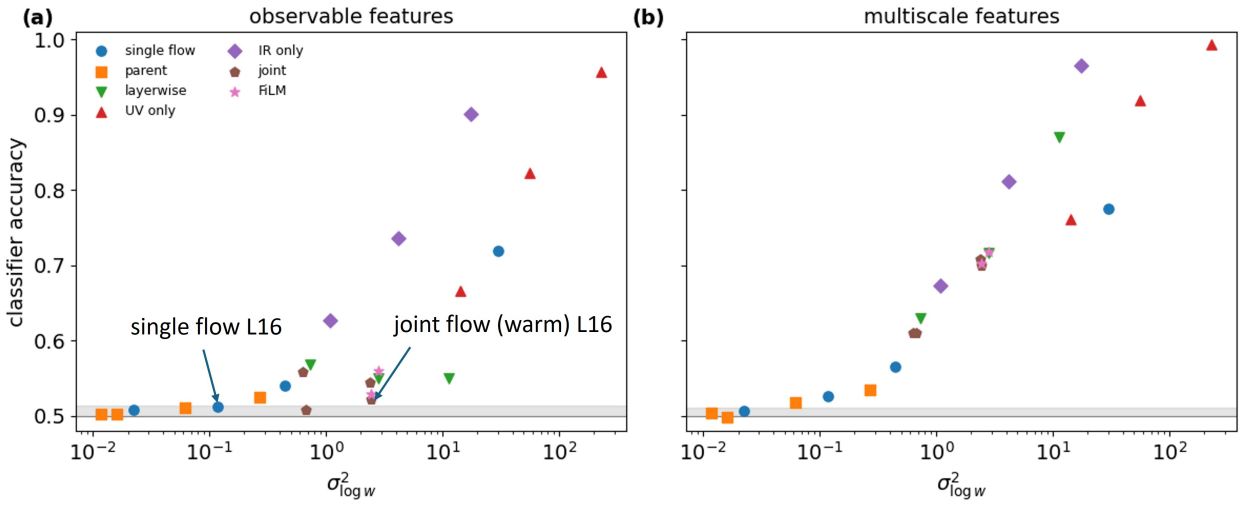


Figure 7: Log-weight variance plotted against the two classifier diagnostics, with the arms in Section 6 and their parent checkpoints shown as separate points. Panel (a) uses $\mathrm{C2ST}_{\mathrm{obs}}$ and panel (b) uses $\mathrm{C2ST}_{\mathrm{MS}}$. Chance accuracy at 0.5 is marked by the horizontal line; each shaded band shows the corresponding training-seed span from Appendix I.1. Similar observable-map accuracies can coexist with very different variances, whereas the multiscale map reveals much of that separation. Both panels remain feature-dependent diagnostics rather than distances between the full distributions.

percentage points in the mean absolute magnetization error, compared with a seed spread of 3.0 points. Likewise, $\mathrm{C2ST}_{\mathrm{obs}}$ changes by at most 0.010 against a spread of 0.014, $\mathrm{C2ST}_{\mathrm{MS}}$ by at most 0.004 against a spread of 0.011, and the Binder error by 1.01 points against a spread of 4.0 points.

Efficiency changes from 0.050 to 0.017 and from 0.115 to 0.096, so direct multiplicative modulation improves neither schedule. Although the first change is large, its finite-sample estimate is tail-sensitive at a log-weight variance near 2.8, while the variance-based lift cost remains unchanged.

State-dependent expansion and contraction are already represented in the additive form through the features of Eq. 45 and can be influenced indirectly by the conditioning term through the trajectory. FiLM adds a direct condition-dependent gain, but the null result shows only that this pathway does not reduce the measured cost under the tested optimization, not that the additive endpoint family has condition-independent scale.

Neither tested change removes the deficit at these lattice sizes. Per created coefficient, a stage costs 36 times the per-site rate of a single flow under the matched update and batch schedule, with only part of the efficiency gap recovered by unfreezing the parent and none by direct multiplicative conditioning. The stage also has 4.6–5.6 times as many trainable parameters and uses 1.59–1.75 times as much peak memory as the direct flow, so these raw counts exclude simple parameter shortage but not limitations in effective expressivity or optimization within the coordinate-restricted arrangement.

After these variations, the residual remains associated with the implemented factorized arrangement, in which the fine distribution is represented by a detail marginal followed by a coarse conditional. The UV transformation moves the $3L^2$ detail coefficients while fixing the L^2 coarse coefficients, and the IR transformation then moves the coarse coefficients while fixing the details; a direct flow, by contrast, couples all sites within one transformation.

Under the matched update and batch schedule, better measured performance is obtained by the direct flow from $L = 8$ through $L = 32$. A factor-34 reduction in lift cost would be required to match its per-site rate, but several features, including depth, ordering, locality and optimization, have not been varied, as listed in Section 7.7. The observed association should therefore not be interpreted as identifying factorization alone as the cause.

7.6 What the Measured Cost Implies for a Hierarchy

A hierarchy can begin at any lattice coarser than its target. This choice determines both the number of stages and the number of coefficients they create: reaching $L = 64$ from $L_0 = 4$ requires the four stages $4 \to 8 \to 16 \to 32 \to 64$, whereas beginning at $L_0 = 32$ requires only the $32 \to 64$ stage.

As reported in Section 7.3, the Binder-shape witness for the required stage correction decreases by a factor of 4.3 across the three transitions, while δ^2 remains stable within 4.1% over the same range. This component of the correction therefore does not predict measured stage cost, so the following analysis instead examines the arithmetic obtained by extending the measured rate through a hierarchy.

A composed sampler carries the variance of its base model and the successive variance increments defined by Eq. 54. Because every target site is either inherited from the base or created by a stage, the stages create $L^2 - L_0^2$ coefficients in total. Let δ^2_{base} denote the base-model value of Eq. 54, and let $\delta^2_{\mathrm{stage}}$ denote the common stage value. The total variance is

$$\sigma^2_{\log w} = \delta^2_{\mathrm{base}} L_0^2 + \delta^2_{\mathrm{stage}} \left(L^2 - L_0^2\right) = \delta^2_{\mathrm{stage}} L^2 + \left(\delta^2_{\mathrm{base}} - \delta^2_{\mathrm{stage}}\right) L_0^2. \tag{55}$$

A single $\delta^2_{\mathrm{stage}}$ stands for every stage in this projection. Although this approximation is consistent with the three separately trained one-step lifts, no composed multistage hierarchy or $32 \to 64$ lift was evaluated, so it is an extrapolation rather than a measured endpoint.

Within this constant-rate projection, the preferred starting lattice is determined by the sign of $\delta^2_{\mathrm{base}} - \delta^2_{\mathrm{stage}}$, with a coarser starting lattice favored only when a stage is cheaper per created coefficient than a base model is per site. Since the measured values are 4.3×10^{-4} for the base and 1.48×10^{-2} for a stage, the projection favors the finest starting lattice that can be trained.

If the two rates are equal, the composed variance becomes $\delta^2_{\mathrm{base}} L^2$, the same rate as a direct flow with the corresponding per-site cost at the target. A lifting stage therefore matches this reference at every L_0 once its per-coefficient cost reaches the flow's per-site cost, which is a factor of 34 below the measured stage cost.

Table 9: Projection of the two terms of Eq. 55 for every available starting lattice at a target of $L = 64$. The base term takes δ^2_{base} from the row of Table 4 at that L_0, and from Appendix H at $L_0 = 4$. The lifting term takes $\delta^2_{\mathrm{stage}} = 1.48 \times 10^{-2}$ from Table 5, representative of the three measured one-step transitions. The unmeasured $32 \to 64$ lift is assumed to carry the same rate.

L_0	stages	$L^2 - L_0^2$	$\delta^2_{\mathrm{base}} L_0^2$	$\delta^2_{\mathrm{stage}} \left(L^2 - L_0^2\right)$	$\sigma^2_{\log w}$
4	4	4080	0.012	60.6	60.6
8	3	4032	0.022	59.8	59.9
16	2	3840	0.118	57.0	57.1
32	1	3072	0.444	45.6	46.0

As shown in Table 9, the base contributes less than one percent of every total, so the two terms in Eq. 55 are not yet competitive. The projected total variance ranges from 46.0 to 60.6, at either of which the ESS of an importance-weighted estimator is effectively zero for every feasible generated batch.

The reference above is a per-DOF rate measured from $L = 8$ through $L = 32$. It does not continue to $L = 64$, where mode collapse is reported in Section 6.1; it therefore sets the scale of the required improvement rather than a target that either construction has been shown to reach at $L = 64$. Untested modifications are listed in Section 7.7.

7.7 Limitations

Scope of the claims. The bound in Eq. 28 applies only to refinements that preserve the inherited coarse coordinates, whereas the IR transformation used here changes those coordinates and is therefore not subject to it. A sufficiently rich bijective stage could, in principle, make the layerwise construction exact. A separate ordering applies to the reverse-KL objective: because the layerwise parameter setting is feasible under joint optimization, the globally attainable joint optimum can be no worse, although this statement neither orders the attained values of ESS/N nor proves a strict gap without additional capacity assumptions.

Protocol and diagnostics. Each layerwise lift uses a finite pool drawn from a frozen parent q_L. Modes absent from this pool cannot be restored by reweighting, and moving q_L toward π_L does not remove the physical mismatch between π_L and π^{IR}_{2L}. Consequently, δ^2 is interpreted as the incremental cost of a frozen-parent lift rather than as a universal end-to-end ranking across training schedules. Checkpoints are selected by smoothed training loss without a holdout, so selection is not optimized for the reported importance-sampling efficiency, although the same rule is used for every arm.

Both C2ST results are feature dependent. The observable map was specified in the original evaluation protocol; the multiscale map was added afterward to test whether its conclusion survives a broader symmetry-aware projection. Although reference-versus-reference calibration and the fixed feature count guard against a trivial dimensional effect, neither classifier is Bayes optimal on raw configurations, so the two values are reported together as a sensitivity analysis.

A systematic comparison with equivariant raw-field classifiers and alternative spectral or real-space projections is left for future work.

Under Eq. 54, the stage is charged for the $3L^2$ coefficients it creates. Charging all $4L^2$ coordinates acted on by the stage would multiply δ^2 by three quarters and change the ratio in Section 6.2 from 36 to 27, without changing the design requirement of Section 7.6. At the smallest reported ESS/N, only about 29 configurations carry appreciable weight, so those values are reported to two significant figures. Frozen-parent comparisons at $L \geq 16$ rely primarily on δ^2, while ESS/N is retained as the endpoint efficiency diagnostic.

Numerical and reference uncertainty. The reported density is obtained by numerical integration rather than from the exact Jacobian of the finite-step map, but doubling the integration grid changes ESS/N by a relative 10^{-4} or less, and raw-sample observables provide a density-independent check, excluding integration error as an explanation for the measured deficit. The effective reference-sample count decreases from 9155 at $L = 4$ to 2057 at $L = 64$, and the $L = 64$ reference Binder cumulant is below an independent result at the same coupling, which may indicate a small offset from the critical line. This reservation affects the $L = 64$ direct-flow diagnosis and the final row of Table 1, but not the central comparisons from $L = 8$ through $L = 32$.

Generality of the construction. The experiments concern one two-dimensional scalar theory at one near-critical coupling and use a fixed dyadic Haar splitting, with each stage containing one UV–IR alternation in the order $q(d)q(c \mid d)$ and velocity fields based on full-lattice convolutions. Repeated alternations, the reverse order $q(c)q(d \mid c)$, bounded-receptive-field kernels, other wavelets, additional dimensions and group-valued fields all remain outside the present scope. The residual cost is therefore associated with the implemented arrangement rather than evidence of a universal limitation of multiscale or factorized samplers.

8 Conclusion

The question addressed here is whether a modular coarse-to-fine sampler can be extended one frozen stage at a time when every resolution is supervised by its own physical action, a choice that creates a consistency obstruction because the distribution supplied by a trained parent is generally not the coarse marginal required by the next target. A lower bound shows that no refinement preserving the inherited coordinates can remove this discrepancy, which must instead be avoided through end-to-end training against the finest action or an explicit correction of the inherited marginal. The second option preserves modularity, but its correction cost must be included when scaling is assessed.

That cost was isolated with a density-evaluable Haar lifting stage that assigns new detail coefficients and inherited coarse coefficients to separate UV and IR transformations. For a frozen parent, δ^2 measures the incremental log-weight variance per newly created coefficient, and its agreement with the paired covariance check together with its stability across three transitions supports its use as a stage-local diagnostic in this protocol. Rather than a universal end-to-end score across training schedules, it answers the question needed for a modular hierarchy: how much additional importance-weight variance does one frozen-parent lift introduce?

The main empirical result is that this cost does not decrease over the measured range, remaining 31–44 times the per-site cost of a direct flow trained under the same update and batch schedule. The ablations identify reconciliation of the inherited coarse distribution as the dominant difficulty: holding the IR transformation at the identity increases lift cost by a factor of about 20, whereas holding the learned UV transformation there increases it by about 1.5. Joint optimization improves the selected magnetization observables but only modestly improves multiscale distinguishability and recovers little of the efficiency gap, confirming that agreement in a small observable projection does not diagnose the layerwise model's density-ratio quality.

Rather than establishing a universal limitation of multiscale sampling, these findings establish a concrete requirement for the tested modular strategy: the cost per created coefficient must approach the per-site cost of a direct flow, whereas the measured stage remains approximately a factor of 34 above that target. Changing the starting lattice does not remove the requirement because every newly created coefficient accumulates stage cost. Adjacent-scale marginal mismatch should therefore be measured before stacking a hierarchy, and any mechanism introduced to correct it should be evaluated separately from detail generation. The resulting bound, density-evaluable construction and stage-local diagnostic together identify where modular scaling fails and quantify the improvement required for it to become competitive.

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A Notation and Abbreviations

Table 10: Notation grouped by the objects being described: the lattice target, the multiresolution coordinates, the invertible transformations and the evaluation quantities.

symbol	meaning	defined in
Target and lattice		
L	linear extent of the lattice, which carries L^2 sites	Sec. 2.1
Λ_L	site set of the $L \times L$ periodic lattice	Sec. 2.1
L_0	lattice size of the coarsest level of a hierarchy	Sec. 4
ϕ	field configuration, one real value per site	Sec. 2.1
m^2, λ	bare mass squared and quartic coupling	Eq. 2
$S_L(\phi)$	ϕ^4 action at lattice size L	Eq. 2
Z_L	partition function, the intractable normalizer	Sec. 2.1
π_L	normalized target density, proportional to $e^{-S_L(\phi)}$	Sec. 2.1
$\mathbb{Z}_2$	global sign symmetry $\phi \mapsto -\phi$	Sec. 2.1
Multiresolution coordinates		
x_ℓ	configuration at resolution level ℓ	Sec. 2.3.1
R_ℓ	coarse-graining map at level ℓ	Sec. 2.3.1
E_ℓ	map from a parent configuration to inherited coarse coordinates	Sec. 2.3.2
$\pi_\ell^{\mathrm{IR}}, \pi_\ell^{\mathrm{UV}}$	target coarse marginal and detail conditional	Sec. 2.3.1
$\overline{\pi}_{\ell-1}$	coarse law inherited from the adjacent parent target	Sec. 2.3.2
c	coarse, or infrared (IR), Haar coefficient field of L^2 values	Eq. 35
$d_{\mathrm{H}}, d_{\mathrm{V}}, d_{\mathrm{D}}$	detail, or ultraviolet (UV), Haar coefficient fields, $3L^2$ values	Eq. 35
c_0	coarse field inherited from the parent, before the IR transformation	Eq. 70
c_1	coarse field after the IR transformation	Eq. 72
d_0, d_1	detail fields before and after the UV transformation	Eq. 32
W_{IRUV}	orthonormal Haar analysis transformation	Eq. 35
$\pi_{\mathrm{free}}(\cdot \mid c)$	conditional free-theory source for the details	Eq. 40
Transformations and model densities		
t	flow time, running over $[0, 1]$	Sec. 2.2.1
p_0	tractable source density of a direct flow	Sec. 2.2
q_θ, q_L	learned model density and its lattice-indexed form	Sec. 2.2
v_θ	velocity field with parameters θ	Eq. 46
$T_{\mathrm{UV}}, T_{\mathrm{IR}}$	invertible UV and IR transformations	Eq. 32
$\Delta \log q_{\mathrm{UV}}, \Delta \log q_{\mathrm{IR}}$	UV and IR log-density changes	Eq. 50
q_{2L}	density of the composed stage at the fine lattice size	Eq. 50
D_4	dihedral symmetry group of the square lattice	Sec. 4.4
Evaluation		
N	number of generated samples	Sec. 5.3
w	importance weight, $e^{-S_L(\phi)}/q_L(\phi)$	Sec. 2.2.2
$\sigma^2_{\log w}$	variance of the log importance weights at one lattice size	Sec. 2.2.2
ESS/N	effective sample size fraction	Sec. 5.3
δ^2	incremental log-weight-variance cost per created degree of freedom	Eq. 54
$\sigma^2_{\log w}/L^2$	log-weight variance per site of a direct single-scale flow	Eq. 54
S	action per site, $S_L(\phi)/L^2$	Sec. 5.3
m	magnetization	Eq. 6
$\langle m \rangle$	ensemble-mean magnetization and sector-balance diagnostic	Sec. 5.3
$\langle \lvert m \rvert \rangle$	mean absolute magnetization, the average distance of m from zero	Sec. 5.3
χ	sector-width susceptibility, $L^2 \operatorname{Var}(\lvert m \rvert)$	Sec. 5.3
$\sqrt{\chi}/L$	width of the magnetization inside one sector	Sec. 5.3
U_4	Binder cumulant	Eq. 7
ΔU_4	blocked-minus-bare Binder-cumulant difference	Sec. 2.4
r_σ	single-site standard-deviation ratio	Sec. 2.4
r_{m^2}	mean-squared-magnetization ratio	Sec. 2.4
$\mathrm{C2ST}_{\mathrm{obs}}$	classifier test on S, m and $\lvert m \rvert$	Sec. 5.3
$\mathrm{C2ST}_{\mathrm{MS}}$	classifier test on multiscale local and spectral features	Sec. 5.3

Table 11: Abbreviations, with the point at which each is introduced in the text.

abbreviation	expansion	introduced in
AdamW	Adam with decoupled weight decay	Sec. 5.1
C2ST	classifier two-sample test	Sec. 5.3
CNF	continuous normalizing flow	Sec. 2.2
DOF	degree of freedom	Sec. 2.1
ESS	effective sample size	Sec. 5.3
FFJORD	free-form continuous dynamics for scalable reversible generative models	Sec. 2.2
FiLM	feature-wise linear modulation	Sec. 6.5
GiB	gibibyte	Sec. 6.1
IR	infrared, denoting the coarse Haar coefficient field	Eq. 35
KL	Kullback-Leibler	Sec. 2.2.2
MCMC	Markov chain Monte Carlo	Sec. 5.3
MsIGN	multiscale invertible generative network	Sec. 3.3
ODE	ordinary differential equation	Sec. 2.2
QCD	quantum chromodynamics	Sec. 3.4
RG	renormalization group	Sec. 3.4
RiGCS	renormalization-informed generative critical sampler	Sec. 3.4
RKL	reverse Kullback-Leibler	Sec. 2.2.2
RK4	fourth-order Runge-Kutta	Sec. 4.4
SRFlow	super-resolution flow	Sec. 3.3
UV	ultraviolet, denoting the detail Haar coefficient fields	Eq. 35

B Haar Coordinates and the Conditional Free Theory

This appendix fixes the transform convention used throughout, records the fixed Jacobian contribution to the stage density, and derives both the conditional Gaussian detail source and its efficient momentum-space construction.

B.1 Transform Convention

Within each 2×2 block, the orthonormal Haar transform is represented by the matrix in Section 4.1. Let W_{IRUV} denote its action on the full lattice. Orthonormality gives

$$W_{\mathrm{IRUV}}^{\top} W_{\mathrm{IRUV}} = I, \qquad |\det W_{\mathrm{IRUV}}| = 1. \tag{56}$$

Thus, no log-density contribution is produced by the coordinate change between the fine field and (c, d), which motivates using an orthonormal rather than a merely invertible transformation. Every density term in Eq. 50 can then be attributed to a learned transformation or to the parent embedding.

B.2 Parent Embedding and Its Jacobian

Before a fine field is formed, the parent is embedded in the low-pass coefficients by treating each parent value as constant over its 2×2 block. The block therefore contains (X_L, X_L, X_L, X_L), and the low-pass transformation returns

$$c_0 = \tfrac{1}{2}\left(X_L + X_L + X_L + X_L\right) = 2X_L, \tag{57}$$

which fixes the factor of two and makes the map diagonal, with entry two on each of its L^2 components. Therefore,

$$\log \left| \det \frac{\partial c_0}{\partial X_L} \right| = L^2 \log 2. \tag{58}$$

This is the only fixed Jacobian term in the stage density, with every other contribution obtained from an integrated divergence.

B.3 Conditional Free Theory

Let the reference free action be $S_{\mathrm{free}}(X) = \frac{1}{2} X^{\top} A X$, where A is the periodic precision matrix of a positive-mass Gaussian field. After transformation to Haar coefficients, the precision is partitioned into low-pass and detail blocks:

$$K = W_{\mathrm{IRUV}} A W_{\mathrm{IRUV}}^{\top} = \begin{pmatrix} K_{cc} & K_{cd} \\ K_{dc} & K_{dd} \end{pmatrix}. \tag{59}$$

For a Gaussian specified by a precision matrix, conditioning on the low-pass block gives

$$\pi_{\mathrm{free}}(d \mid c) = \mathcal{N}\left(-K_{dd}^{-1} K_{dc} c,\ K_{dd}^{-1}\right), \tag{60}$$

with normalized log density

$$\log \pi_{\mathrm{free}}(d \mid c) = \tfrac{1}{2} \log\det K_{dd} - \tfrac{3L^2}{2} \log(2\pi) - \tfrac{1}{2}(d - \mu_c)^{\top} K_{dd} (d - \mu_c), \qquad \mu_c = -K_{dd}^{-1} K_{dc} c. \tag{61}$$

Two properties of this construction are used in Section 4.2: the conditional mean is linear in c, and the conditional covariance is independent of c. At the free-theory level, the dependence of the detail law on the coarse field is therefore a pure displacement that additive conditioning can represent exactly.

B.4 Momentum-Space Construction

A dense construction of K, which would require a $4L^2 \times 4L^2$ matrix, is unnecessary because A is translation invariant and the Haar blocks tile the lattice regularly. Consequently, K is block-circulant on the coarse lattice and becomes block diagonal in its Fourier basis, with each block $K(k)$ having size 4×4. Its symbol is obtained by applying the fine-lattice precision to one coefficient impulse per channel and transforming the result back to coefficient space, requiring only four precision applications and no dense matrix; Hermiticity is imposed explicitly to remove rounding asymmetry.

At each coarse momentum, the detail block $K_{dd}(k)$ has size 3×3. Sampling and density evaluation therefore reduce to independent 3×3 Cholesky solves, giving a total cost of $O(L^2 \log L)$ dominated by the transforms. Before factorization, a small multiple of the identity is added to $K_{dd}(k)$ to protect positive definiteness against rounding error, while using the same modified precision for sampling and evaluation keeps the resulting density exactly normalized.

C Analytic Divergence of the Velocity Field

This appendix derives the closed-form divergence used by every transformation in Section 4, identifies the contribution from conditioning, and shows that a multiplicative gain preserves the construction.

C.1 Velocity and Features

Let x denote a transported coefficient field with channels indexed by a and sites by s, and let y denote the fixed conditioning field. Site-wise features are formed from a linear term and trainable sine frequencies:

$$F_p(u) = \sum_f P_{pf}\,\varphi_f(u), \qquad \varphi_f(u) = \begin{cases} \sin(\omega_f u), & f < n, \\ u, & f = n, \end{cases}, \tag{62}$$

so that both F_p and its derivative are available in closed form,

$$F_p'(u) = \sum_{f<n} P_{pf}\,\omega_f \cos(\omega_f u) + P_{pn}. \tag{63}$$

Let $*$ denote circular convolution over the periodic coefficient lattice, and obtain the time-dependent kernels K and H by contracting a Fourier time basis with the orbit-tied kernel banks, giving the velocity

$$v_a(x,t;y)[s] = \sum_{b,p} \big(K_{abp}(t) * F_p(x_b)\big)[s] + \sum_{b,p} \big(H_{abp}(t) * F_p(y_b)\big)[s]. \tag{64}$$

C.2 The Trace

Only the diagonal derivatives $\partial v_a[s]/\partial x_a[s]$ are required for the divergence, and under differentiation of Eq. 64, the convolution contributes its kernel coefficient at zero displacement, the only term coupling the output at site s to the input at that site. Using the feature derivative in Eq. 63, while noting that the second sum is independent of x and therefore vanishes, gives

$$\nabla_x \cdot v = \sum_a \sum_s \sum_p K_{aap}(t)[0]\; F_p'\big(x_a[s]\big). \tag{65}$$

This form has three useful properties: it is exact and therefore introduces neither Hutchinson sampling nor additional estimator variance into the loss; D_4 orbit tying determines which parameters populate K; and it applies to both local and full-lattice kernels because each has a well-defined zero-displacement coefficient.

C.3 Multiplicative Conditioning

The transported contribution can also be scaled by the condition:

$$v_a(x,t;y)[s] = g_a(y,t)[s] \sum_{b,p} \big(K_{abp}(t) * F_p(x_b)\big)[s] + \sum_{b,p} \big(H_{abp}(t) * F_p(y_b)\big)[s]. \tag{66}$$

The derivation is unchanged because the gain $g_a(y,t)$ depends on the condition and time, but not on the transported field. The trace becomes

$$\nabla_x \cdot v = \sum_a \sum_s g_a(y,t)[s] \sum_p K_{aap}(t)[0]\; F_p'\big(x_a[s]\big), \tag{67}$$

Only one factor is added to the existing contraction, allowing the condition to enter the divergence and directly modulate contraction at a given state, as motivated by Appendix E.2. When initialized at unity, the gain makes Eq. 66 a strict generalization of Eq. 64.

C.4 What the Integration Retains

The state features in Eq. 62 and their derivatives in Eq. 63 vary along the trajectory and must therefore be retained at every solver stage. By contrast, the orbit-expanded kernel banks depend only on the parameters, and the condition features only on the fixed conditioning field, so both can be computed once per integration rather than at each of the $4n$ right-hand-side evaluations in an n-step RK4 solve. This rearrangement leaves numerical values unchanged while reducing peak memory by 45%–59% in the configurations used here, with larger savings on larger coefficient lattices.

D Bijectivity of the Stage Map

The UV and IR updates are described in Section 4.5 as sequential triangular maps whose determinant contributions multiply. Proving this claim together with the exact change-of-variables formula in Eq. 50 establishes that one stage is a bijection from the parent field and detail source to the complete fine configuration space.

Although the result is elementary, it requires each transformation to define a complete flow and therefore requires a global Lipschitz bound on the velocity field. This bound follows from the specific features in Eq. 45 but would not hold for an arbitrary unbounded feature choice, so it is stated explicitly.

D.1 Statement

Fix a stage with parent size L. Let $N_c = L^2$ be the number of coarse coefficients and $N_d = 3L^2$ the number of detail coefficients, whose sum $N_c + N_d = 4L^2$ equals the number of fine-lattice sites. The stage map is

$$\Phi : \mathbb{R}^{N_c} \times \mathbb{R}^{N_d} \longrightarrow \mathbb{R}^{4L^2}, \qquad \Phi(X_L, d_0) = X_{2L}, \tag{68}$$

and carries a parent field and detail-source draw to the generated fine configuration. Apart from the necessary equality of dimensions, the remaining conditions are established below without assuming any particular origin for X_L. Following the generation order of Section 4.3, Φ factorizes as

$$\Phi = W_{\mathrm{IRUV}}^{-1} \circ R \circ U \circ E, \tag{69}$$

with

$$E : (X_L, d_0) \mapsto (c_0, d_0), \qquad c_0 = W_{\mathrm{IR}}(X_L) = 2X_L, \tag{70}$$

$$U : (c_0, d_0) \mapsto (c_0, d_1), \qquad d_1 = T_{\mathrm{UV}}(d_0; c_0), \tag{71}$$

$$R : (c_0, d_1) \mapsto (c_1, d_1), \qquad c_1 = T_{\mathrm{IR}}(c_0; d_1), \tag{72}$$

where $T_{\mathrm{UV}}(\cdot\,; c_0)$ and $T_{\mathrm{IR}}(\cdot\,; d_1)$ denote the time-one maps of the two ODEs of Section 4.3, each with its condition held fixed.

Proposition 1 *Φ is a continuously differentiable bijection of $\mathbb{R}^{4L^2}$ onto itself. Its Jacobian is nonsingular everywhere and satisfies*

$$\log|\det J_\Phi| = L^2 \log 2 + \log|\det J_{\mathrm{UV}}| + \log|\det J_{\mathrm{IR}}|\,, \tag{73}$$

with J_{UV} and J_{IR} the fixed-conditioning Jacobians of Eq. 50. No cross term between the two transports appears.

D.2 The Two Linear Maps

Lemma 1 *E is a linear isomorphism with $\log|\det J_E| = L^2 \log 2$.*

Proof. In the coordinates of Eq. 68, E has the block-diagonal matrix $\mathrm{diag}(2I_{N_c}, I_{N_d})$, whose determinant $2^{N_c} = 2^{L^2}$ is nonzero, so E is invertible and the stated logarithm follows. □

Lemma 2 *W_{IRUV} is orthogonal, so W_{IRUV}^{-1} is a linear isomorphism with $\log|\det J_{W^{-1}}| = 0$.*

Proof. W_{IRUV} acts block-diagonally over the L^2 Kadanoff blocks of Section 4.1, and within a block its matrix is $H = \frac{1}{2}M$ with

$$M = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{74}$$

Each row of M has squared norm 4, and distinct rows are orthogonal. Therefore, $MM^\top = 4I_4$ and $HH^\top = I_4$, making H orthogonal with $|\det H| = 1$. A block-diagonal matrix of orthogonal blocks remains orthogonal, and interleaving those blocks into the $2L \times 2L$ array only permutes coordinates with absolute determinant one. Appendix B gives the same calculation when fixing the convention. □

Because H is also symmetric, $H^{-1} = H$, which is why analysis and synthesis carry identical coefficients.

D.3 The Transports Are Complete Flows

Lemma 3 *The velocity field of Eq. 46 is globally Lipschitz in its transported argument, uniformly in $t \in [0,1]$ and in the condition. That is, there exists $K < \infty$ with*

$$\|v(x,t;y) - v(x',t;y)\| \le K \|x - x'\| \qquad \text{for all } x, x' \in \mathbb{R}^n,\ t \in [0,1],\ y. \tag{75}$$

Proof. Write $v(x,t;y) = G_\theta(t) * F(x) + H_\theta(t) * F(y)$. The second term does not depend on x and therefore contributes nothing.

The features in Eq. 45 act site-wise, and every raw feature has a bounded derivative: for the sine channels, $|\partial_u \sin(\omega_k u)| \le |\omega_k|$, while for the linear channel, $\partial_u u = 1$. Since the trainable frequencies ω_k are finite and the projection P is linear, F is globally Lipschitz, with $L_F \le \|P\| \max(1, \max_k |\omega_k|)$.

Periodic convolution with $G_\theta(t)$ is a bounded linear operator. Its kernel bank is a finite combination of Fourier time features and is continuous on the compact interval $[0,1]$. Thus, $C_G := \sup_{t\in[0,1]} \|G_\theta(t)\| < \infty$. Tying coefficients across D_4 orbits imposes only a linear constraint and does not change this bound, so the product $K = C_G L_F$ is sufficient. □

The chosen features make this bound global rather than local, which is not a generic property of CNFs. Features with faster-than-linear growth, such as polynomial or exponential functions, would generally provide only a local Lipschitz constant and would require finite-time blow-up to be excluded separately before defining the time-one map.

Lemma 4 *Let $v : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$ be continuous in t, continuously differentiable in x, and globally K-Lipschitz in x. The time-one map φ_1 of $\dot{x} = v(x,t)$ is then defined on all of $\mathbb{R}^n$ and is bijective. A C^1 inverse is obtained by integrating the reversed field $\tilde{v}(x,s) := -v(x, 1-s)$.*

Proof. Existence and uniqueness. Picard–Lindelöf gives a unique maximal solution from every initial condition.

No blow-up. The Lipschitz bound gives linear growth, $\|v(x,t)\| \le M + K\|x\|$ with $M := \sup_t \|v(0,t)\| < \infty$. Since $\frac{d}{dt}\|x(t)\| \le \|\dot{x}(t)\|$, Grönwall's inequality yields

$$\|x(t)\| \le \big(\|x(0)\| + \tfrac{M}{K}\big)\, e^{Kt} - \tfrac{M}{K}, \tag{76}$$

for $K > 0$. When $K = 0$, the corresponding bound is $\|x(t)\| \le \|x(0)\| + Mt$; both bounds are finite for $t \in [0,1]$, so every maximal solution reaches $t = 1$ and φ_1 is defined everywhere.

Injectivity. Suppose $\varphi_1(a) = \varphi_1(b) = z$. The reversed field $\tilde{v}$ is continuous in s and K-Lipschitz in x, making the solution of $\dot{x} = \tilde{v}(x,s)$ started at z unique; since both a and b equal that solution at $s = 1$, they are equal.

Surjectivity. Given $z \in \mathbb{R}^n$, integrate $\tilde{v}$ from z over $s \in [0,1]$; the same growth bound guarantees existence, and the endpoint w satisfies $\varphi_1(w) = z$.

Differentiability in the initial condition follows from smoothness of v in x. This smoothness is supplied by the sine and linear channels, and the Jacobian is consequently nonsingular. □

With the condition held fixed, Lemmas 3 and 4 apply to both transformations, so $T_{\mathrm{UV}}(\cdot\,; c_0)$ is a diffeomorphism of $\mathbb{R}^{N_d}$ for every c_0, and $T_{\mathrm{IR}}(\cdot\,; d_1)$ is a diffeomorphism of $\mathbb{R}^{N_c}$ for every d_1.

D.4 Triangularity of the Lift and the Relaxation

Lemma 5 *U and R are bijections of $\mathbb{R}^{N_c} \times \mathbb{R}^{N_d}$ with*

$$\det J_U = \det J_{\mathrm{UV}}, \qquad \det J_R = \det J_{\mathrm{IR}}. \tag{77}$$

Proof. In the block coordinates (c, d),

$$J_U = \begin{pmatrix} I_{N_c} & 0 \\ \partial d_1/\partial c_0 & \partial d_1/\partial d_0 \end{pmatrix}, \qquad J_R = \begin{pmatrix} \partial c_1/\partial c_0 & \partial c_1/\partial d_1 \\ 0 & I_{N_d} \end{pmatrix}. \tag{78}$$

J_U is block lower triangular, and J_R is block upper triangular. Each determinant is therefore the product of the determinants of its diagonal blocks, giving the stated identities, while Lemma 4 ensures that the relevant blocks are nonsingular. Explicit inverses are $(c_0, d_1) \mapsto (c_0, T_{\mathrm{UV}}^{-1}(d_1; c_0))$ and $(c_1, d_1) \mapsto (T_{\mathrm{IR}}^{-1}(c_1; d_1), d_1)$. □

The two triangular structures have opposite orientations: during the lift, the details are transformed using the *inherited* coarse coefficients, whereas during relaxation, the coarse coefficients are transformed using the *generated* details. This alternating coupling layer has off-diagonal blocks $\partial d_1/\partial c_0$ and $\partial c_1/\partial d_1$ that do not enter either determinant but are generally nonzero and carry communication between the two transformations.

Proof of Proposition 1. By Eq. 69, Φ is the composition of E, U, R and W_{IRUV}^{-1}. These maps are bijective by Lemmas 1, 5 and 2, so their composition is bijective, while the chain rule preserves continuous differentiability and nonsingularity of the Jacobian. Multiplying the four determinants and taking logarithms gives Eq. 73, with zero contribution from the Haar factor and $L^2 \log 2$ from the embedding.

□

For each transport, the instantaneous change-of-variables identity gives $\log|\det J_{\mathrm{UV}}| = \int_0^1 \nabla_d \cdot v_{\mathrm{UV}}\, dt$, with an analogous expression for the IR transport; combining these identities with Eq. 73 reproduces Eq. 50. The source $q_L(X_L)\,\pi_{\mathrm{free}}(d_0 \mid c_0)$ is a valid joint density on $\mathbb{R}^{N_c} \times \mathbb{R}^{N_d}$ even though the law of d_0 depends on X_L through $c_0 = 2X_L$.

D.5 Consequences and Scope of the Proof

Source of the parent field. X_L appears in Eq. 68 only as an argument, and its distribution is not used in the proof. Lemma 1 concerns a linear map, Lemmas 3 and 4 concern only the stage velocity, and Lemma 2 concerns a fixed orthogonal matrix. Proposition 1 therefore holds regardless of how the parent field is produced, whether by a frozen checkpoint, an unfrozen level in joint training, a reference Markov chain or an arbitrary configuration. The density identity requires only that q_L be a valid density at the parameter setting in use, whether or not those parameters are being updated.

The same reasoning applies to the full hierarchy. Under joint training, the composite map begins at the source of the coarsest model, passes through that model and every lifting stage, and ends at the finest configuration. Each coarse model is itself a CNF with the velocity field of Section 4.4. Lemmas 3 and 4 therefore apply without modification, making the composition bijective, while dimensions agree because $L_0^2 + \sum_\ell 3L_\ell^2$ equals the number of finest-lattice sites.

Invertibility is retained under joint training, but quantities defined relative to q_L change interpretation: first, the increment in Eq. 54 would subtract a parent variance no longer measured against the parent's own target; second, the inherited marginal of Section 2.3.2 is no longer fixed because its level can move; and third, the bound in Eq. 28, which applies to refinements preserving the coarse coordinates, does not apply when those coordinates are transformed twice. For these reasons, Section 6.4 reports absolute efficiency rather than δ^2 for the joint arms.

Full support. The map Φ covers all of $\mathbb{R}^{4L^2}$, and the source density is positive everywhere, so $q_{2L} > 0$ over the full configuration space, the importance weights of Section 2.2.2 are finite, and the self-normalized ESS is well defined. No region of π_{2L} is assigned zero model probability.

Joint character of the stage density. Proposition 1 gives a bijection onto the full fine configuration space, so q_{2L} is a joint density over all $4L^2$ DOF rather than a product of independently fitted factors. Since W_{IRUV} is orthogonal, this density can be re-expressed under any other linear splitting, such as one between retained and newly resolved sites, without changing its value. The corresponding marginal does change: in particular, the coarse marginal of q_{2L} describes block averages rather than field values on a sublattice, and Section 2.3.2 concerns the former.

Relation to expressiveness. Nothing above depends on the model being accurate: Proposition 1 establishes that Eq. 50 is a valid density, not that q_{2L} is close to π_{2L}. The effect of additive conditioning on the attainable endpoint family, discussed in Section 4.4 and tested in Section 6.5, is untouched by the bijectivity result.

Discretization of the transport. The transports are approximated with fixed-step RK4. The step map $x \mapsto x + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ has Jacobian $I + O(hK)$ and is locally nonsingular when the step size is sufficiently small relative to the trained Lipschitz constant, although local nonsingularity does not establish global invertibility of the discrete map. Proposition 1 applies exactly to the continuous flow, of which the implementation is a numerical approximation, and the reported log density is likewise the continuous-time divergence integral rather than the exact log determinant of the finite-step map. Appendix I.3 bounds the discrepancy empirically by refining the grid at fixed random draws.

E Further Diagnostics of the Adjacent-Resolution Mismatch

E.1 Amplitude of the Correction

In Section 2.4, ΔU_4 is used to show that the required coarse correction is not a pure rescaling, while the two amplitude diagnostics in Table 1 describe the component that rescaling can capture. They are recorded for completeness and support no main-text claim.

The single-site standard deviation ratio r_σ increases monotonically. Its values are 0.8427, 0.8510, 0.8563 and 0.8575, with successive increments of 0.0083, 0.0053 and 0.0012, consistently with a plateau near 0.857. However, four sizes cannot establish convergence or transferability of the amplitude correction, and a bootstrap over blocks of 50 consecutive configurations estimates the error on the last value as ± 0.0016. This estimate remains stable across a twenty-five-fold range of block lengths and exceeds the final increment of 0.0012.

Similar behavior is observed for the mean squared magnetization ratio r_{m^2} over the first three transitions, but the final value of 0.7128 breaks the trend; this change can be traced to the turnover of the bare Binder cumulant discussed in Section 2.4 and is not pursued.

Neither quantity supports a reported result; instead, they show that one constant per level can absorb the overall amplitude, while ΔU_4 isolates the component that resists a near-linear transformation.

E.2 Dependence of the Coarse Conditional on the Details

Each stage first generates the detail coefficients and then transforms the coarse field conditionally on them, requiring the conditional law of c given d to be represented. This law is characterized on the $L = 16$ reference ensemble through its location and scale, both evaluated at fixed local detail energy $\|d_x\|^2$.

Symmetry fixes the location because the action in Eq. 2 is even under $\phi \mapsto -\phi$, whereas c_x is odd and $\|d_x\|^2$ is even, so the conditional mean of c_x vanishes at every detail energy. Its conditional standard deviation does depend on detail energy, and when reported over octiles in Table 12 and normalized to an unconditional value of one, it contracts monotonically from 1.092 to 0.835, a modulation of roughly one fifth. Randomly permuting the detail energies recovers a control value of 1.00 in every bin, while the same unpermuted pattern appears at $L = 8$ and $L = 32$.

Table 12: Scale of the coarse conditional at fixed local detail energy, measured on the $L = 16$ reference ensemble and binned into octiles. Because $\mathbb{Z}_2$ symmetry forces the conditional mean to vanish, only the variation in scale is reported; randomly permuting the detail energies gives a ratio of 1.00 in every bin.

octile	1	2	3	4	5	6	7	8
mean detail energy	0.098	0.235	0.366	0.510	0.680	0.900	1.225	2.064
relative scale of c	1.092	1.071	1.048	1.029	1.004	0.969	0.928	0.835

No trained model is involved in this measurement, which supports no reported result but identifies the conditioning required by the target; Section 6.5 tests the conditioning supplied by the construction against this requirement.

F The Parent Law and Its Endpoints

In Section 4.6, training parents are drawn uniformly from a finite pool; the more general law interpolates between uniform model sampling and target-corrected selection. This appendix identifies the corresponding population law and its limitations.

F.1 Population Limit

Let $X^{(1)}, \dots, X^{(N)}$ be independent draws from the frozen parent q_L, and let

$$w(X) = \frac{\exp\left[-S_L(X)\right]}{q_L(X)}, \qquad \ell_i = \log w\Big(X^{(i)}\Big), \tag{79}$$

so that the selection probabilities read

$$\rho_i = (1-\beta)\frac{1}{N} + \beta\,\frac{w\big(X^{(i)}\big)^\alpha}{\sum_j w\big(X^{(j)}\big)^\alpha}. \tag{80}$$

For the law of a single drawn parent, the uniform component contributes q_L directly, while for the importance component, the law of large numbers gives

$$\frac{1}{N}\sum_j w\Big(X^{(j)}\Big)^{\alpha} \;\longrightarrow\; \mathbb{E}_{q_L}[w^{\alpha}] = \int q_L^{1-\alpha}\exp\left[-\alpha S_L\right], \tag{81}$$

so that in the large-pool limit the selected parent is distributed as

$$r_{\alpha,\beta} = (1-\beta)\,q_L + \beta\,\frac{q_L^{1-\alpha}\pi_L^{\alpha}}{Z_\alpha}, \qquad Z_\alpha = \int q_L^{1-\alpha}\pi_L^{\alpha}, \tag{82}$$

where the powers of the intractable normalization of π_L are absorbed into Z_α and therefore never need to be evaluated.

F.2 Endpoints

The cases of interest are contained in Eq. 82. At $\beta = 0$, the parent law is exactly q_L, making the stage loss the ordinary reverse KL of the composed model and recovering the plain layerwise scheme. At $\beta = 1$ and $\alpha = 1$, the law is π_L, so the stage is trained on parents from the coarse physical target rather than the model. For $\beta = 1$ and $0 < \alpha < 1$, the geometric bridge $q_L^{1-\alpha}\pi_L^{\alpha}/Z_\alpha$ tempers the correction and limits the weight assigned to any single pool member, while an intermediate β mixes a corrected component into an otherwise model-faithful pool. At $\alpha = 0$, q_L is recovered for any β, so the two parameters are not independent at this endpoint.

The minibatch loss is left unweighted, so the correction enters only once through selection and the loss remains interpretable as training on samples from $r_{\alpha,\beta}$ rather than as a reweighted loss on samples from q_L.

F.3 What the Parent Law Cannot Do

Two limitations follow directly from Eq. 82.

First, $r_{\alpha,\beta}$ is absolutely continuous with respect to q_L for every α and β, so every member of the family assigns zero density wherever q_L does. A region never generated by the parent model, including a missing mode, therefore cannot be recovered by reweighting a finite pool; either parameter can only improve the weighting of regions already covered.

Second, although interpolation between q_L and π_L can address the approximation error between the frozen model and the coarse physical target, it cannot address the discrepancy between π_L and π_{2L}^{IR} established in Section 2.3.2. This discrepancy is a property of the two physical theories and remains even for an exact parent, so the fully corrected endpoint moves the parent toward the wrong target for the refinement task. Accordingly, the parent law is a controllable training choice rather than a remedy for the central problem.

G Reference Ensembles

Reference ensembles are generated by hybrid Monte Carlo with a leapfrog integrator, using the action in Eq. 2, the couplings in Eq. 4, and seed 42 at every lattice size. Table 13 records the chain parameters, and Table 14 reports the observables.

Table 13: Reference ensemble generation, with thermalization, thinning and τ_{int} given in leapfrog trajectories. The integrated autocorrelation time is measured for $|m|$ because the reported observables are even in the magnetization, and N_{eff} denotes the implied number of effectively independent configurations.

L	therm.	stored	thin	step	leapfrog	acc.	τ_{int}	N_{eff}
4	5000	10000	10	0.10	10	0.979	5.5	9155
8	5000	10000	20	0.10	10	0.959	11.4	8739
16	5000	10000	30	0.10	10	0.917	23.0	6534
32	5000	10000	40	0.10	10	0.836	76.0	2631
64	8000	20000	80	0.10	10	0.679	388.9	2057

The $L = 64$ chain is thermalized for 8000 trajectories and stores 20000 configurations, compared with 5000 and 10000, respectively, at the smaller sizes. Although the $L = 64$ autocorrelation time is 5.1 times that at $L = 32$, its thinning interval is twice as long, yielding 2057 effectively independent configurations compared with 2631 at $L = 32$; the thermalization length is approximately $20\,\tau_{\mathrm{int}}$.

Table 14: Observables of the reference ensembles under the convention of Eq. 2, against whose matching lattice-size row every relative error in Sections 6 and 7 is evaluated. The magnetization is defined in Eq. 6, the susceptibility and Binder cumulant in Section 5.3 and Eq. 7, respectively, and S denotes the action per site.

L	$\langle \lvert m \rvert \rangle$	$\langle m^2 \rangle$	χ	$\sqrt{\chi}/L$	U_4	$\langle S \rangle \pm \mathrm{sd}(S)$
4	0.7624	0.7268	2.329	0.3815	0.4533	0.3540 ± 0.1780
8	0.6771	0.5396	5.195	0.2849	0.5175	0.3772 ± 0.0961
16	0.5999	0.4084	12.415	0.2202	0.5540	0.3879 ± 0.0503
32	0.5284	0.3127	34.369	0.1832	0.5689	0.3932 ± 0.0269
64	0.4414	0.2229	115.198	0.1677	0.5540	0.3964 ± 0.0142

Two additional $L = 64$ chains are used only as consistency checks. The first contains 10^4 configurations with a thinning interval of 80, while the second contains 10^3 configurations with a thinning interval of 400 after 20000 thermalization trajectories. The three chains agree to four figures in $\langle \lvert m \rvert \rangle$ and give Binder cumulants of 0.5552, 0.5540 and 0.5503. No reported result relies on the available $L = 128$ ensemble, whose thermalization is shorter than two autocorrelation times and which contains only about 150 effectively independent configurations.

H Single-Scale Flows at Larger Batch

The single-scale flows reported here were trained with the largest batch size permitted at each lattice size: 256, 192, 128, 64 and 32 for $L = 4$ through $L = 64$, respectively, compared with 64 for every lifting stage. Consequently, the single-scale flows see between 1.7 and 3.0 times as many configurations as a stage at the same target size; their quality is reported in Table 15.

The checkpoints at $L = 4$, $L = 8$ and $L = 16$ are used as the parents of every reported lifting stage and in every frozen-transport, joint and FiLM arm. The per-created-coefficient increment δ^2 of Eq. 54 subtracts the parent's log-weight variance, and although a paired covariance can affect the interpretation of this difference, Appendix I.2 finds it negligible for the reported frozen-parent arms.

Table 15: Single-scale flows trained with the largest batch permitted at each lattice size. The budget is given as the number of updates times the batch size. The $L = 4$, $L = 8$ and $L = 16$ rows provide the parent checkpoints for every reported stage. Errors and both C2ST diagnostics are evaluated against the reference ensembles in Appendix G. The sign of every error is retained.

L	budget	ESS/N	δ^2	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST_{obs}}/\mathrm{C2ST_{MS}}$	peak memory
4	3000×256	0.985	7.4×10^{-4}	-2.6%	-0.009	-0.8%	-0.3%	-0.1%	0.502/0.504	2.14 GiB
8	6000×192	0.985	2.50×10^{-4}	$+0.3\%$	$+0.005$	-1.5%	$+1.6\%$	-1.5%	0.503/0.498	6.37 GiB
16	8000×128	0.940	2.40×10^{-4}	-0.8%	$+0.007$	$+0.9\%$	$+3.4\%$	-0.2%	0.511/0.519	15.60 GiB
32	10000×64	0.744	2.64×10^{-4}	$+0.4\%$	-0.000	-4.8%	$+5.4\%$	-2.6%	0.525/0.535	31.56 GiB
64	10000×32	0.019	7.39×10^{-3}	-1.8%	$+0.002$	$+29.5\%$	-64.0%	$+18.0\%$	0.719/0.776	64.33 GiB

The per-site log-weight variances are 2.50×10^{-4}, 2.40×10^{-4} and 2.64×10^{-4} at $L = 8$, 16 and 32, respectively, while ESS/N decreases from 0.985 to 0.744 across these sizes. Below $L = 64$, the action error remains within one percent, the mean absolute magnetization error ranges from -0.8% to -4.8%, and the width error grows from -0.3% at $L = 4$ to $+5.4\%$ at $L = 32$, while $\mathrm{C2ST_{obs}}$ increases from 0.502 to 0.525 and $\mathrm{C2ST_{MS}}$ from 0.504 to 0.535. At $L = 64$, the mode collapse described in Section 6.1 gives corresponding classifier values of 0.719 and 0.776 and a width error of -64.0%.

I Reproducibility and Protocol Checks

I.1 Seed Variance

The $4 \to 8$ layerwise lift was repeated four times with the configuration, budget, parent checkpoint and evaluation seed held fixed, while only the random seed used for stage training was varied.

The variation caused by the training seed alone is reported in Table 16. The full ranges of ESS/N and the per-created-coefficient cost δ^2 are 6.3% and 6.7% of their means, respectively, while the action error spans 0.43 percentage points.

Table 16: Four replicates of one transition that differ only in the training seed. Errors and both C2ST diagnostics are evaluated against the reference ensembles in Appendix G. The sign of every error is retained. A standard deviation estimated from four samples has an uncertainty of approximately 40%. Therefore, the variation quoted for each column is the full range spanned by the four runs.

seed	ESS/N	$\delta^2/10^{-2}$	S error	$\langle m \rangle$	$\langle \lvert m \rvert \rangle$ error	width error	U_4 error	$\mathrm{C2ST_{obs}}/\mathrm{C2ST_{MS}}$
4567	0.460	1.51	$+1.9\%$	-0.006	-13.2%	$+3.5\%$	-11.8%	0.568/0.630
4568	0.476	1.51	$+1.4\%$	-0.007	-10.1%	$+3.7\%$	-8.6%	0.554/0.622
4569	0.450	1.51	$+1.6\%$	-0.007	-10.8%	$+3.7\%$	-9.2%	0.557/0.627
4570	0.479	1.62	$+1.5\%$	-0.007	-11.5%	$+6.5\%$	-12.6%	0.561/0.634

The mean absolute magnetization, Binder and width errors span 3.0, 4.0 and 2.9 percentage points, respectively; the $\mathrm{C2ST_{obs}}$ and $\mathrm{C2ST_{MS}}$ spans are 0.014 and 0.011, and the mean magnetization spans 0.0002.

I.2 Paired Covariance Check for the Incremental Lift Cost

The change in log-weight variance is accounted for exactly by Eq. 54, but a difference of variances is not itself the variance of the error introduced by a lift. Because the released samples retain each fine draw together with the parent draw from which it was generated, define

$$W_{\mathrm{in}} = \log w_L(X_L), \qquad W_{\mathrm{out}} = \log w_{2L}(X_{2L}), \qquad R_{L\to 2L} = W_{\mathrm{out}} - W_{\mathrm{in}}. \tag{83}$$

The variance increment then decomposes exactly as

$$\delta^2 = \frac{\mathrm{Var}(R_{L\to 2L})}{3L^2} + \frac{2\,\mathrm{Cov}(W_{\mathrm{in}}, R_{L\to 2L})}{3L^2}. \tag{84}$$

The first term is nonnegative and directly measures the lift's additional log-weight fluctuation relative to its parent, while the second records its correlation with the inherited error.

Table 17: Paired decomposition of the incremental cost for the three full frozen-parent lifts. The covariance share is the second term of Eq. 84 divided by δ^2.

transition	δ^2	$\mathrm{Var}(R_{L\to 2L})/(3L^2)$	covariance share
$4 \to 8$	0.015143	0.015198	-0.37%
$8 \to 16$	0.014539	0.014492	$+0.33\%$
$16 \to 32$	0.014840	0.014793	$+0.32\%$

A spread of 4.8% is obtained for the paired residual rate across the three transitions, compared with 4.1% for the variance difference, and its ratios to the direct-flow per-site rate are 43.9, 31.4 and 34.1, leaving the interpretation and all headline comparisons unchanged. Repeating the decomposition for every frozen-transport arm, seed replicate and the layerwise FiLM arm keeps the magnitude of the covariance share below 1.22% in all cases. This small value is an empirical property of the accurate frozen parents and trained lifts used here rather than a general identity for differences of log-weight variances.

I.3 Checkpoint Selection and Convergence Checks

The integration grid is checked by regenerating configurations on the native grid and at twice its resolution while holding every random draw fixed, so any difference can be attributed to integration error. Doubling the number of steps from 80 to 160 changes the layerwise-stage samples by a root mean square of 3.4×10^{-5} against a field scale of 0.89, changes the log density by 5.0×10^{-6} per site, and changes ESS/N by a relative 9.7×10^{-5}. For a jointly optimized stage that composes two trained transformations, the corresponding values are 8.0×10^{-5}, 9.0×10^{-6} and 2.2×10^{-5}.

J Hardware

Runs were dispatched to a shared heterogeneous pool, with the scheduler determining the device assigned to each run. The available accelerators included the NVIDIA A100 80GB, L40S, RTX 3090, RTX 3090 Ti, RTX 4070 Ti SUPER, RTX 4090, RTX 5060 Ti, RTX 5070 Ti, RTX PRO 5000 Blackwell and RTX PRO 6000 Blackwell.